

Group 5 Final Report

School of Management, Tianjin Normal University Course Completion Report

Semester of Class:	Second Semester, 2024–2025 Academic Year	Course Name:	Information System Development Methods and Tools
Name:		School:	School of Management
Class:		Student ID:	

Reference Scoring Criteria: (Proposition teachers may formulate scoring criteria independently according to the course assessment requirements)

Project	Indicator	Score	Rating
Completeness of Development Document	includes all necessary content for the entire process, such as planning, analysis, design, implementation, operation, and maintenance	10	
Normativity of Development Document	The graphics meet the specifications, the text description is clear and fluent, the format is aesthetically pleasing, and there are no low-level errors such as typos	10	
Source Code	The source code is complete and clear, with the key functional parts containing necessary comments, enabling immediate explanation of the code's functionality.	5	
Operational Reliability	can operate normally all functions, without any errors or crashes occurring in between, taking into account safety factors	5	
Functional Integrity	The functions implemented by the final system are no less than those planned in the analysis phase, with all necessary functions fully implemented, and it includes at least 5 non - administrative functions	5	
Interactivity	The interface is aesthetically pleasing, interactive operations are easy to learn, and the cost for users to learn and use is low	5	

Successfully Delivered	The system meets user needs, and users have signed to approve that the system can address actual pain points	20	
Peer Review	Each group will draw lots to conduct targeted mutual evaluations, and the evaluations must be carried out seriously and responsibly.	10	
Additional Score for Merit Evaluation Report	The content of the report is smooth and complete, the demonstration logic is clear, and scores are assigned according to different levels of excellence evaluation	25	
Role Bonus	Assign scores based on the different roles students play in the course and their daily performance	5	
Total		100	
Comment Marker Year Month Day			

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Development Report of Campus Book Recycling Second-hand Book Circulation System

1 Introduction

Against the backdrop of the continuous expansion of higher education and the iterative updates of textbooks, the demand for the circulation of campus second-hand books has become increasingly prominent. Traditional offline second-hand book trading models, such as flea markets and posting small advertisements, have problems such as scattered trading information, low communication efficiency, and limited trading time and space, making it difficult to meet students' needs for convenient, efficient, and secure second-hand book transactions. Meanwhile, a large number of idle textbooks and extracurricular books pile up, not only causing resource waste but also increasing students' economic burden. With the rapid development of Internet technology, online trading platforms, with their convenience, efficiency, and wide reach, have provided a new solution for the circulation of campus second-hand books. Developing a dedicated campus second-hand book circulation system can not only integrate the idle book resources on campus, build a centralized and standardized trading platform, realize the recycling of books, and practice the concept of green environmental protection; it can also provide teachers and students with convenient book purchase channels, effectively reduce the cost of book purchase, and meet diverse reading needs. This development report of the campus second-hand book circulation system will elaborate on the system's development background, objectives, functional design, technical solutions, and development process, aiming to demonstrate how the system addresses the existing problems in campus second-hand book transactions through technical means and provides strong support for campus resource sharing and knowledge transfer.

1.1 Background Introduction

Based on the 2023 industry Statistical Data, the scale of China's second-hand book market has already exceeded 100 billion yuan and maintained an annual growth rate of over 18%¹. Among numerous segmented markets, the college market occupies a pivotal position. Taking the relevant survey of Guizhou colleges as an example, among the college population in Guizhou, 33% of colleges have already established second-hand book trading markets, and the penetration rate of second-hand book purchases among students has reached 30%². Take a certain university for instance; within just one semester, the demand for second-hand books among its students was approximately 30,000 copies, which fully demonstrates the high-frequency demand characteristics of the campus market for second-hand books³.

In the traditional campus second-hand book trading model, transactions mainly rely on offline stalls or scattered communities. However, this model has many drawbacks, such as information asymmetry caused by poor information circulation, low transaction efficiency caused by cumbersome transaction processes, and the lack of an effective trust guarantee mechanism, etc. 45. According to the research results from Guizhou universities

- 1 https://www.sohu.com/a/828629195_122004016
- 2 <https://m.book118.com/html/2024/0415/8123055025006057.shtm>
- 3 <https://m.book118.com/html/2024/0415/8123055025006057.shtm>
- 4 <https://blog.csdn.net/sheji1051/article/details/137068773>
- 5 <https://blog.csdn.net/wuzhou101/article/details/146342168>

Results show that only 38% of students' universities have established second-hand book trading platforms, and these platforms generally suffer from limited book collections and a single variety of books, ultimately resulting in a low success rate of second-hand book transactions.

From 2012 to 2024, the number of Chinese literature on campus second-hand books steadily increased in the early stage (2012 - 2019), reflecting the gradual accumulation of academic attention; it remained stable from 2020 to 2021, reached a significant peak in 2022, and declined from 2023 to 2024. The Period-on-period Ratio fluctuated synchronously, with the high growth in 2022 corresponding to the peak of literature volume, and the subsequent decline indicating a phased cooling of the popularity. Overall, it reflects that the research on campus second-hand books is significantly affected by external factors, with fluctuations in development, indicating that the level of attention to campus second-hand books is not low.



Figure 1 Academic Attention to Campus Second-hand Books

From 2011 to 2025, the volume of literature on campus second-hand books showed a fluctuating trajectory: it grew rapidly from 2017 to 2019, reflecting the attention campus second-hand books received during this period; it remained at a high level from 2020 to 2022 and gradually declined after 2023. The Period-on-period Ratio fluctuated in tandem, with negative growth from 2024 to 2025 after high growth in 2018, indicating that the academic research enthusiasm for campus second-hand books may enter an adjustment period, requiring new

drivers such as technological innovation to restart growth to meet the demand for campus resource recycling.



Figure 2 Academic Dissemination of Campus Second-hand Books

From the perspective of academic attention trends, the demand for campus second-hand book circulation has long existed and has the potential for fluctuating growth. When constructing the circulation system, it is necessary to draw on the law of research popularity fluctuations, respond quickly during peak attention periods, optimize the functions of online platforms, achieve accurate search, convenient transactions, and information transparency, integrate offline channels, utilize technological means, and use big data recommendation to solve the problems of transaction trust and resource matching. At the same time, combined with the needs of environmental protection and economic burden reduction, promote the use of the system, facilitate the efficient circulation of second-hand books, contribute to the recycling of campus resources and the economic burden reduction of students, and make the system a link between supply and demand.

Therefore, every year as graduation season approaches, college graduates have a large number of used books to dispose of. Meanwhile, freshmen enrolling each year have a huge demand for textbooks and other books, so the campus second-hand book market has great potential. However, the time when graduates and freshmen are on campus does not overlap, and the misalignment of the on-campus time between sellers and buyers leads to low efficiency in second-hand book transactions. Therefore, a trading platform that overcomes the time misalignment and connects graduates with freshmen is necessary.

During their college years, students use a large number of textbooks and books, most of which have low utilization rates after graduation. As a result, a significant amount of book resources are wasted each year when graduates leave school. Taking the School of Ecosystem and Environment of Inner Mongolia University as an example, a questionnaire-based sampling survey of idle book resources was conducted, with a total of 200 questionnaires distributed and 166 valid questionnaires recovered. The main results of the questionnaire are as follows: the idle rate of textbooks for public courses such as English, mathematics, and politics is 33%, the idle rate of textbooks for professional courses is 28%, and the idle rate of reference books is relatively high, reaching 74%. Taking conservative data, the overall idle rate of books is 30%. The

value of idle books of students in the School of Ecosystem and Environment was estimated, and the results are shown in Table 1. The value of idle books of graduates from the entire school each year is 210,000 yuan. Based on this estimate, the value of idle book resources in the hands of graduates from Inner Mongolia University each year is as high as 3.6 million yuan. Recycling these books can not only generate economic benefits but also have social value, theoretically reducing carbon dioxide emissions by 120 tons, saving energy by 1.6×10^5 kJ, and saving 800 tons of fresh water [9]. In conclusion, the recycling and utilization of a large number of idle books on campus can save resources and have both economic and social benefits.

Number of graduates (persons/year)	Per capita number of books (volumes)	Per capita book value (yuan)	Book Idle Rate (%)	Idle value of books (yuan)
180	120	4000	30%	210000

Table 1 Estimation of the Value of Idle Books in the School of Ecosystem and Environment, Inner Mongolia University

1.2 Purpose of System Development

This system aims to build a closed-loop trading model of "online platform + offline QR code". Its core objectives include addressing the pain points of traditional transactions, integrating scattered transaction information through the online platform, generating QR codes to enable convenient offline transactions, breaking through time and space constraints, and resolving issues such as information asymmetry and cumbersome processes. It also activates the value of idle resources, provides low-cost textbooks for price-sensitive freshmen, enables quick monetization of used books for graduating students, and avoids "hanging

It addresses the dilemma of "single slow-moving books turning into waste paper" while meeting the precise acquisition needs of academic users for scarce foreign books. By practicing the concept of green environmental protection, through the carbon credit visualization module, an environmental protection bill is generated upon completion of the transaction to quantify the environmental benefits of second-hand book trading and promote the circular economy model of "trading in old for new".

1.3 Significance of System Development

The system uses technological means to convert idle books into circulable resources. Taking Inner Mongolia University as an example, if 3.6 million yuan worth of idle books across the university are recycled, theoretically 120 tons of carbon emissions can be reduced, equivalent to planting 6,700 trees, combining both ecological benefits and resource conservation value. The

carbon credit exchange mechanism, such as donating for credits, can also motivate students to participate in environmental protection actions, integrating the concept of a "green campus" into daily transactions. According to calculations, each circulated textbook can reduce 0.3 kg of carbon emissions. If the system covers 30% of second-hand book transactions across the university, it can be equivalent to planting 600 trees annually. The "carbon credit" mechanism quantifies environmental protection actions into visible benefits, driving students to shift from "passive consumption" to "active environmental protection." For price-sensitive freshmen, the system's book recommendation page provides price information for books, facilitating comparison during purchase, which can reduce the cost of textbooks for a single semester and save expenses compared to buying new books; for graduates, it can accelerate the circulation of old books and reduce transportation costs. Additionally, it provides a channel for scarce resources to meet research needs, facilitating the sharing of academic resources. The system adopts a B/S architecture to implement full-process digital management of "online transaction - offline storage and retrieval," reducing development costs while improving transaction efficiency. The AI review (OCR identification of ISBN to prevent piracy) and intelligent inspection and guarantee modules address the challenge of quality verification for second-hand books, filling the technological gap in campus trading platforms.

2 Market Research

2.1 System Development Goal 1: Instant Transaction Community Function with Zero Trust Cost

2.1.1 Core Pain Points of Target Users

An analytical mind map centered around the core pain points of college students, which starts from three aspects: full lifecycle management of books, incentive mechanism for environmental protection behavior, and optimized management of campus space:



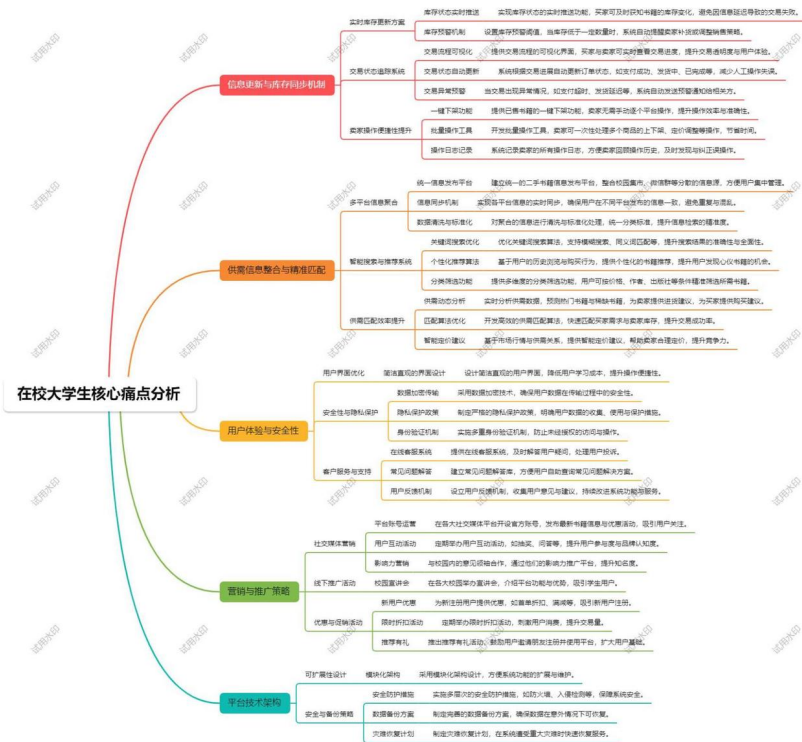
Analysis of Core Pain Points of College Students 1 in Figure 3

2.2 System Development Goal 2: Optimize the efficiency of supply-demand matching, reduce transaction delay and information redundancy functions

2.2.1 Core Pain Points of Target Users

The mind map of the analysis of the core pain points of college students focuses on multiple scenarios of campus life and breaks down pain points and response directions from five major dimensions. Overall, it follows the process of "identifying pain points in college students' campus life - breaking down scenario requirements - providing optimization

Transforms "pathways" into logic, covering scenarios such as learning, life, and social interaction, facilitating the iteration of campus services and management, and enhancing the student experience.



Analysis of Core Pain Points of College Students 2

2.3 User Interview

Regarding the initial design and selection of system development goals, we conducted interviews with users:

Interview Questions	User Feedback
Do you think it would motivate you to participate more actively if the system generated an environmental bill showing the amount of paper saved and carbon emissions reduced after each second-hand book transaction or book donation?	If the system generates an environmental protection bill showing the amount of paper saved and carbon emissions reduced after each second-hand book transaction or book donation is completed, I think this will definitely motivate me to participate more actively.
If the seller uploads a book video, after the AI automatically reviews its quality	If the seller uploads a book video, after the AI automatically reviews its quality

When the transaction amount is immediately credited, as a buyer, would you be worried that the book does not match the description?	The transaction amount will be credited immediately, so as a buyer, I probably won't be too worried about the actual condition of the book not matching the description. AI is really powerful now and can clearly assess the condition of the book.
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	Moreover, since there is a video, I can carefully check whether the book is damaged or not before placing an order. If the Content Moderation passes, the quality of the manual is basically okay. Even if there are really minor issues, I think the platform must have follow-up handling measures and won't let me suffer, so I'm not too worried that the received book will be different from what's in the video.
As a seller, would you be willing to pay a small commission to the platform?	If I were a seller, I might be willing to pay a small commission to the platform. After all, the platform has provided us with such good trading opportunities and helped us sell our books. With the platform, I don't have to look for buyers everywhere on my own, and I can also enjoy services such as instant payment and Content Moderation. Moreover, if the commission is not too much, deducting a little from the book sales revenue seems quite reasonable to me. It's like paying for convenience, which can make transactions smoother, and I'm more willing to sell books on this platform.
Based on our introduction and the above questions, do you think our system is helpful to you, and is it feasible?	Based on your introduction and these questions, I think this system is extremely helpful to me. Usually, buying new books is too expensive, and it's hard to find reliable channels for second-hand books. This system can help me easily find the second-hand books I want and also allow me to sell the books I no longer need, which is both cost-saving and environmentally friendly. From an implementation perspective, I think it is completely feasible.

Table 2 User Interview Form

2.4 Observation Method

In the research on campus second-hand book trading channels, the field observation method was used to deeply analyze the operational logic and actual bottlenecks of the existing model. During the research on the on-campus flea market, it was observed that during the graduation season, there was a dense flow of people in front of the stalls, where students negotiated prices on-site based on the condition and price of textbooks, and the characteristics

of instant transactions were relatively prominent. However, during non-event periods, the site vacancy rate was high, and the daily trading demand was difficult to meet. Visiting bookstores revealed that although fixed shelves could display various textbooks, the yellowed price tags indicated that some books had not been updated for a long time, and the store clerks' explanations of the differences between different versions of textbooks were vague, exposing problems in price and category management

The lag in management; when observing the Xianyu College section online, searching for "XX University textbooks" yields over a thousand pieces of information, with shipping fees sometimes even higher than the book price, and complaints of "version mismatch" appearing in the comment section; transaction records from second-hand group chats reveal that over 200 messages flood the chat daily, causing purchase requests to sink to the bottom within half an hour, and the lack of a standardized transaction process leaves after-sales returns and exchanges without clear guidelines. The following is a brief analysis and summary of the advantages and pain points of the four channel types:

Channel Type	Advantages	Pain Point Analysis
Offline Flea Market	Inspect goods in person and conduct instant transactions	Only available during specific periods (such as graduation season), with limited coverage
Used Bookstore	Stable inventory, negotiable price	Price opacity, slow updates, and incomplete product categories
Online platform (Xianyu)	High traffic, convenient operation	Information is cluttered, cross-school transaction shipping costs are high, and after-sales service is not guaranteed
Second-hand Market / Club Group Chat	High trust level	Small scale, information is easily swamped, and lack of standardized processes

Table 3 Observation Method Analysis Table

Based on the above investigation, we have analyzed in detail the issues of the advantages and disadvantages of the above four types of channels:

Channel Type	Advantages	Disadvantages
Offline Flea Market	1. Intuitive experience: Buyers can inspect the condition of the books on-site, browse	1. Time Constrained: Only open during specific time periods (such as the start of the

	through the content, and intuitively assess the quality of the books, thus avoiding disputes arising from inaccurate information. 2. Instant Transaction: The buyer and seller directly communicate and negotiate prices, and can immediately complete the transaction after reaching an agreement, without waiting for logistics and delivery. 3. Environmental protection public welfare: Usually carried out during key periods such as graduation season, it can effectively promote the recycling of used books and practice the concept of environmental protection.	semester or graduation season), unable to meet daily transaction needs. 2. Spatial Limitation: The venue is limited, with a small coverage area, only able to serve students in the vicinity and difficult to attract users from long distances or other campuses. 3. Affected by weather: Flea markets held outdoors are vulnerable to adverse weather conditions, which may lead to the cancellation or interruption of the event. 4. Loose management: lack of unified management, non-standardized transaction processes, absence of after-sales support, and difficulty in resolving disputes.
Used Bookstore	1. Stable inventory: It has a fixed business location and relatively stable book inventory, which can meet the long-term book purchasing needs of students. 2. Professional Screening: The shopkeeper will conduct screening on the recycled books	1. Price opacity: The lack of a unified standard for book pricing may lead to the phenomenon of "quoting prices based on the customer," making it difficult for buyers to judge the reasonableness of prices. 2. Slow update: Slow speed of book recycling and new arrivals,

Channel Type	Advantages	Disadvantages
	Screening ensures that the content of books is complete, without serious damage, and to a certain extent guarantees the quality of books. 3. Negotiable: The buyer and seller can negotiate the price, giving the buyer an opportunity to purchase the desired book at a lower price. 4. After-sales Guarantee: As a fixed business entity, in case of quality issues, buyers can visit the store to negotiate returns or exchanges, and after-sales service is relatively guaranteed.	Popular textbooks or new books may not be quickly available on the shelves, potentially failing to meet students' needs in a timely manner. 3. Limited variety: Inventory mainly consists of common textbooks, with fewer niche or unpopular books, making it difficult to meet personalized reading needs. 4. Operational risks: Impacted by online platforms, some second-hand bookstores are experiencing poor business conditions and face the risk of closure, which affects service stability.

Online Platform units (idle Fish	1. Massive traffic: Relying on the platform's large user base, it has high exposure, can quickly reach a large number of potential buyers and sellers, and expand the scope of transactions. 2. Easy to operate: All operations in the entire process, such as publishing, browsing, communication, and payment, can be completed via the mobile app, without being restricted by time and space. 3. Rich variety: Bringing together sellers from across the country, it offers a wide range of books, meeting the diverse needs of different majors and reading interests. 4. Function Completeness: It has functions such as search, filtering, price comparison, and evaluation, which facilitate users to quickly find their target books and understand product information and seller reputation.	1. Information overload: A vast amount of information dilutes high-quality content, requiring buyers to spend a significant amount of time screening, and there may be false information or exaggerated descriptions. 2. High cross-school costs: When cross-school transactions are involved, logistics shipping fees increase transaction costs, reduce the price advantage of second-hand books, and there is a risk of transportation damage. 3. Lack of trust: Most buyers and sellers are strangers, lacking direct communication and a foundation of trust. This makes it easy for issues such as goods not matching the description, delayed delivery, etc., to occur, and after-sales rights protection is cumbersome. 4. Intense competition: The same book may have multiple sellers, resulting in fierce price wars, squeezing sellers' profit margins and dampening their enthusiasm for transactions.
Second-hand Group Chat	1. High trust level: Group members are mostly classmates, alumni, or people with common interests, who are familiar with each other. Transactions are based on trust relationships, resulting in fewer disputes. 2. Precise matching: Community group chats are often established around specific interests or professional fields, with members having similar needs, enabling quick and precise matching of books to those needs. 3. Efficient communication: Direct communication within the group chat allows for quick inquiries about book details, negotiation of prices, and discussion of transaction methods, resulting in low communication costs and high transaction efficiency.	1. Limited scale: The number of group members is limited, the scope of transactions is narrow, the circulation speed of books is slow, and it is difficult to meet the demand for a large volume of book transactions. 2. Difficulty in information management: Messages in the group chat scroll quickly, and transaction information is easily overwritten by other chat content, causing some members to miss important information. 3. Lack of standardization: There are no standardized trading processes and management systems, trading methods are arbitrary, after-sales service is not guaranteed, and it is prone to conflicts.

	4. Social Interaction: During the transaction process, it can enhance communication among members, strengthen the cohesion of the community, and simultaneously meet the needs for knowledge sharing and social interaction.	4. Dependence on the group owner: The trading order within the group mainly relies on the group owner for maintenance. If the group owner manages poorly or has no time to attend to it, chaos is likely to occur.
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Table 4 Analysis Table of Advantages and Disadvantages of Four Channels

3 System Planning

3.1 Task Time Estimation

Our project tasks consist of ten items, with project initiation taking 0 days, requirement research 3 days, competing product and market analysis 2 days, platform function planning and design 7 days, database architecture setup 5 days, front-end interface development 12 days, back-end function development 15 days, system integration testing and use case execution 10 days, test issue fixing and experience optimization 7 days, platform launch and deployment 5 days, campus promotion and operation 25 days, and the total cycle being

$0 + 3 + 7 + 20 + 10 + 7 + 5 + 25 = 77$ days. The task project has flexibility. If, under ideal conditions, all projects start as early as possible and are completed within the specified number of days, our total task planning and development cycle will be 16 days in March, 30 days in April, and 31 days in May, with all tasks to be completed by June 6 at the latest. The following is the task schedule, including task number, task description, task duration, predecessor task number, and planned start time.

10

Campus Promotion Operation (Value Promotion/Event)

25

9

May 7th

Figure 5 Task Schedule Table

3.2 Draw Gantt Chart and Pert Chart

The following is the PERT Chart, with the Critical Path nodes being 0-1-3-4-6-7-8-9-10:

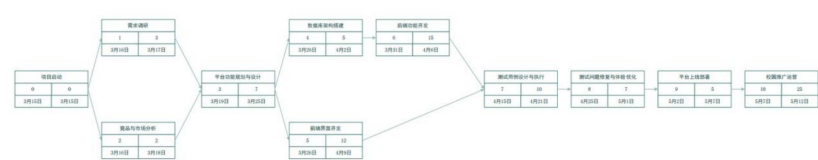


Figure 6 System Planning PERT Chart

The Gantt Chart intuitively displays the overall project timeline and task priorities; helps identify task conflicts or resource idleness, optimizing the scheduling of resources such as manpower and equipment; by comparing planned and actual progress, it promptly detects delay risks and adjusts strategies; provides a unified progress reference for team members, reducing information asymmetry. Below is the Gantt Chart we have drawn:

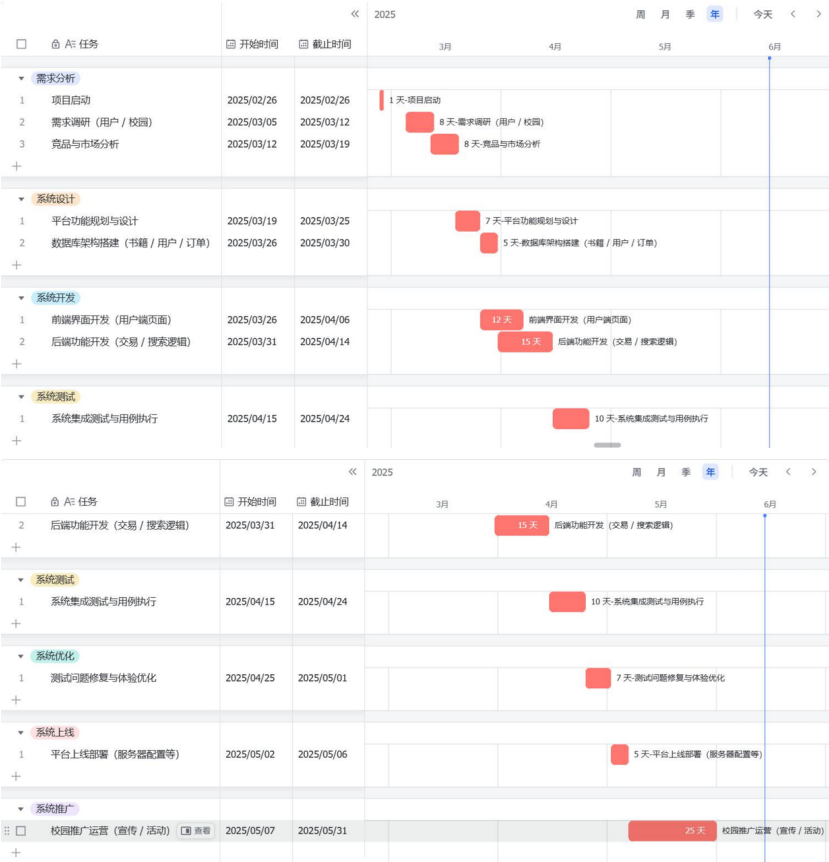


Figure 7 Task Schedule Gantt Chart

3.3 Functions Available in the System

- ① Carbon Credit Visualization: This feature means that every time a second-hand book transaction is completed or a student donates a book, the system will generate an environmental protection bill. On this bill, it clearly states how much paper has been saved and how much carbon emissions have been reduced through this action. In this way, everyone can actually see their contributions to environmental protection, and their enthusiasm for participating in environmental protection will surely skyrocket!
- ②Offline QR code transaction: To address the potential issue of book loss, we will generate a transaction QR code for the transaction between the buyer and seller after the payment is successfully completed. The buyer and seller will conduct an offline transaction, and this QR code can only be used by the buyer

Buyers and sellers can see it, it is relatively secure, and it is extremely convenient. There is no need for manual sorting one by one, which greatly reduces management costs and immediately improves transaction efficiency.

③ Intelligent Inspection Guarantee: In second-hand book transactions, both buyers and sellers are most concerned about the quality of the books. Our system has this function, where the seller uploads a video of the book, and AI will automatically review the actual condition of the book. Once the review is approved, the transaction funds will immediately be transferred to the seller's account. This establishes a standardized inspection process, where buyers don't have to worry about receiving books with damage, missing pages, etc., and sellers don't have to worry about not getting paid after selling their books. The rights and interests of both parties are thus protected, the trust chain is naturally restored, and transactions can proceed more smoothly.

④ Intelligent Recommendation Algorithm: When users use the system to purchase books, there will be recommendations for related books below, providing users with multiple choices and personalized recommendations.

⑤ Multi-dimensional information aggregation: The system will record in detail the version of books, such as the edition number of textbooks, the degree of newness, whether it is 90% new or slightly worn, and other levels. It can quickly locate by subject classification and accurately find the target just like looking up a dictionary.

3.4 Final System Implementation Execution Status

Our team actually carried out the following tasks as required: Requirements Analysis Phase: feasibility analysis, task time estimation, drawing Gantt charts and Pert charts, drawing user portraits and data flow diagrams. System Analysis Phase: drawing UML class diagrams and use case diagrams, drawing empathy maps and logo design. System Overall Design: drawing system design class module diagrams, system structure module diagrams, and system task allocation diagrams, drawing activity diagrams and sequence diagrams; System Detailed Design: detailed use case design and prototype design; System Prototype Design: system design and review analysis report; System Implementation: drawing code flowcharts; user system prototype interaction and feedback; System Testing: Black box, SAST white-box testing; System Delivery and User Feedback.

3.5 Differences between the plan and actual execution

The differences between the plan and actual execution are reflected in use case diagrams, class diagrams, detailed use case design, code flow chart drawing, Black box testing, and SAST white-box testing. Because our group initially defined the function as that of the intelligent module storage intelligent cabinet, but after the subsequent economic feasibility analysis, the

intelligent cabinet function was not so easy to implement, so the subsequent testing would not be so rigorous, which would lead to some differences between the plan and implementation.

4 System Analysis

4.1 Requirements Analysis

Based on the previous market research and initial system planning, we need to conduct a requirements analysis of the system to further determine the feasibility of system implementation. The requirements analysis mainly takes the form of AI interaction, research, and user interviews, and finally summarizes into a core user profile.

Based on the feedback from literature review and user interviews, user pain points cover core dimensions such as economic cost, resource efficiency, sustainable development, and book scarcity:

(1) Economic cost: The pricing of college textbooks and professional books is generally high (the unit price of some science and engineering textbooks exceeds 100 yuan). College students are a group with limited financial resources and find it difficult to afford the cost of new books, resulting in the phenomenon of "buying fewer books or buying pirated books", which affects the learning experience. Moreover, after students graduate or courses end, textbooks are often left idle or sold at low prices to waste recycling stations, causing personal economic losses and resource waste. Sellers cannot efficiently dispose of idle books, and buyers have difficulty obtaining low-cost resources, resulting in "inefficient matching at both ends of supply and demand".

(2) Information Transaction and Efficiency: The traditional campus second-hand book purchase model relies on confession walls, campus bazaars, or the "flea market" during graduation season for transactions, which has problems such as scattered information, limited coverage, and restricted transaction time. Buyers need to spend a lot of time searching for target books, while sellers face the dilemma of "being ignored." Offline private transactions lack a guarantee mechanism, making it prone to disputes such as goods not matching the description and order evasion, and it is difficult to safeguard rights after-sales. Users have low trust in private transactions and worry that their rights cannot be guaranteed, such as the seller disappearing after the buyer pays or the received books having missing pages or excessive notes.

(3) Environmental Protection and Sustainable Development: Universities generate a large amount of idle textbooks every year. If these textbooks are directly discarded or incinerated, it not only wastes wood resources but also increases carbon emissions. According to statistics, on average, a textbook generates 5kg of carbon dioxide from printing to disposal. Students lack environmental awareness or channels, making it difficult to convert idle books into sustainable resources, and campus environmental protection initiatives lack specific carriers.

(4) Research support needs: Academic research often requires supporting datasets, experimental code, and answers to after-class exercises, but finding them on one's own is time-consuming and laborious. Borrowing research books from libraries is sometimes in short supply.

With the help of AI, we finally summarized the users' needs, aiming to eliminate pain points and build a system that meets users' expectations. The main target of the campus second-hand book circulation system is students on campus, and their needs can be summarized as follows:

(1) Economic Sensitivity: The price of used books must be significantly lower than that of new books, and support the price range filtering function.

(2) Precise Book Matching: Quick location is achieved through intelligent search (keywords, ISBN, etc.), multi-dimensional filtering (major, edition, condition, etc.), and popular book recommendations on the homepage. Book details include clear images, complete descriptions (such as damage, note locations, etc.), and seller reputation scores, reducing information asymmetry.

(3) Hopes that textbooks, reference materials, notes, etc. can be efficiently circulated to reduce waste of school resources.

(4) The offline transaction location can be mutually agreed upon by both parties, restricted to within the campus, ensuring safety and convenience.

Based on the above analysis of users' pain points and needs, we have divided users into three categories: price-sensitive, graduation clearance, and academic need. We will also create user personas and empathy maps respectively for further analysis.

1. Price-sensitive users

This type of user is a freshman who has just entered college, with limited monthly living expenses and a semesterly textbook budget not exceeding one-tenth of their living expenses. They highly value cost-effectiveness in textbook consumption. These users are frugal and practical, good at careful planning, and have a high acceptance of new technologies. Facing the high prices of new books and the chaotic second-hand book market, they want to save money by buying second-hand books but are also worried about being scammed. At the beginning of the semester when preparing textbooks and before exam weeks when looking for review materials, they will take active action to find ways to meet their textbook needs. They will hear about the unpleasant experiences of buying pirated books from roommates, learn the experience of choosing second-hand books from upperclassmen and upperclasswomen, and also listen to their parents' reminders about spending money reasonably. When opening a second-hand book app, they are at a loss about how to choose from the vast amount of information. They will look for matching textbooks based on the recommended book lists from upperclassmen and upperclasswomen, silently calculate whether their budget is sufficient when checking the prices of new books in the school bookstore, and mutter about whether

second-hand books are reliable, whether the versions are correct, and whether there are cost-effective channels. To select suitable second-hand textbooks, they will spend a lot of spare time comparing prices on different platforms, carefully reading user reviews to judge the condition of the books, asking upperclassmen and upperclasswomen about textbook versions and taking notes, and communicating with sellers about whether the books are damaged and whether they are the latest edition. However, problems such as the high price of new books exceeding their affordability, the chaotic second-hand book market making it difficult to distinguish the quality and version, and the lack of guidance on book purchases leading to easy pitfalls really trouble this type of user. They hope to find second-hand textbooks with reasonable prices and good quality to save money, and also want to accumulate experience through communication with upperclassmen and upperclasswomen and reading reviews, master the skills to distinguish the quality and version of books, so that buying textbooks can be worry-free and cost-effective, and smoothly advance their studies.



Figure 8 Price-Sensitive User Portrait



Figure 9 Empathy Map of Price-Sensitive Users

2. Graduation Cleanup Users

This type of user is a graduate who hopes to clean up the dormitory as soon as possible. Facing graduation, the old books piled up in the dormitory have become a thorny problem. These users are decisive and efficient, dislike trouble, and want to quickly resolve the disposal of books to reduce the burden of moving when leaving school. Previously, old books listed on the platform couldn't be sold for a long time and could only be treated as waste paper. This time, they want to efficiently recoup their costs and don't want to spend too much time on this matter. One month before graduation, they will use their mobile phones to take photos and batch upload old books to the platform, post book-selling messages in campus groups and alumni groups, and communicate with buyers for transactions. They will also listen to classmates complaining about the difficulty of selling old books, hear freshmen and sophomores seeking to buy books, and hear couriers say they are busy with business. They will worry looking at the books filling the dormitory and the unsold books on the platform, can't help complaining, hoping someone will buy the books, disliking the trouble of selling books one by one, and worrying about logistics. The pain points of this type of user are that old books listed on the platform are difficult to sell and can only be disposed of at a low price, lacking a convenient batch processing method which is time-consuming, and the logistics are tight during the graduation season, making transactions and moving inconvenient. Their expectations are to quickly clear out the books to solve the difficulty of moving, make the old books valuable and obtain economic returns, and also be able to communicate and share with freshmen and sophomores. Under this demand, they prefer the platform's "batch photo upload" and "one-click packaging", tend to choose "recycling and consignment sales", and hope the platform has a "graduation season green channel", "door-to-door book collection", and "donation for points" to help them efficiently dispose of old books and smoothly start their new journey after graduation.



Figure 10 User Portrait of Graduation Cleanup Type



Figure 11 User Empathy Map for Graduation Cleanup

3. Academic demand users

Such users are basically graduate students with heavy research tasks and extremely urgent needs for academic resources. They cannot borrow scarce books from the library and can only turn to the platform. However, the platform has few foreign language resources, and they often encounter sellers who pass off photocopied versions as genuine ones.

The lack of access to resources not only affects reading but may also delay scientific research. The supervisor emphasizes that scientific research should refer to authoritative literature, and classmates have also complained about the unpleasant experience of buying pirated photocopied books. When attending academic lectures, experts recommend cutting-edge works, which makes him deeply aware of the importance of resources. He would feel disappointed when seeing the book he wanted to borrow in the library marked as "borrowed", find few results when searching for foreign-language materials on second-hand platforms, and be full of concerns when verifying the book information from sellers, and can't help complaining about the difficulty of finding professional foreign-language books and the fear of encountering photocopied versions. To find resources, he switches back and forth between academic resource platforms and second-hand book platforms, compares the credibility of sellers and book descriptions, asks sellers for proof of authenticity, and consults supervisors and classmates for channels. Such users are looking forward to a professional and reliable platform that can solve the problems of scarce foreign-language resources and difficulty in distinguishing authenticity, successfully obtain scarce genuine books to advance scientific research, accumulate experience in authenticity verification, and exchange experiences in resource acquisition with peers. By means of "foreign-language book subscription" and "academic exchange community", etc., they can make the scientific research path smoother and meet the pursuit of knowledge authority and efficient acquisition.



Figure 12 Academic Demand User Portrait

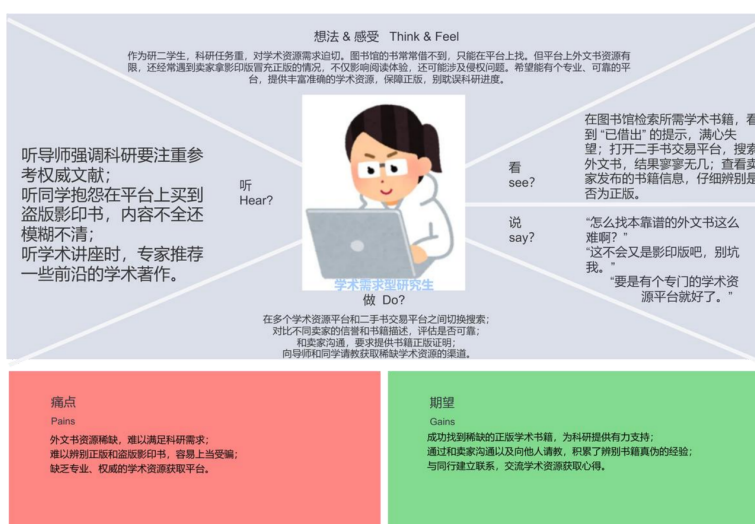


Figure 13 Academic Needs Empathy Map

4.2 System Function Analysis

4.2.1 Functional Analysis

(1) Carbon Credit Visualization Function

In the use case diagram of the carbon credit visualization module, the system encompasses three types of participants: buyers, sellers, and administrators. Centered around the user login scenario, "Login / Registration" serves as the core use case, extending through generalization relationships to two specific implementation methods, "WeChat Login" and "Mobile Login", to adapt to different users' login habits; meanwhile, it supports the "Forgot Password" use case through an extension relationship, providing users with the supplementary function of password retrieval. At the business function level, buyers and sellers can jointly participate in the "Credit Exchange" use case to implement the operation of exchanging carbon credits for gifts; when both parties complete the transaction, it will trigger the "Transaction Completed" use case, which in turn drives the "Carbon Credit Calculation" use case, where the system

calculates the carbon credits generated by the transaction behavior according to the rules, ultimately generating an "Environmental Bill". This use case diagram, with the "Carbon Credit Visualization Module" as its boundary, comprehensively presents the interaction logic between participants and functional use cases, including both the diversified design of the login system and the full process of credit exchange and post-transaction carbon credit processing, providing clear use case logic support for the implementation of system functions.

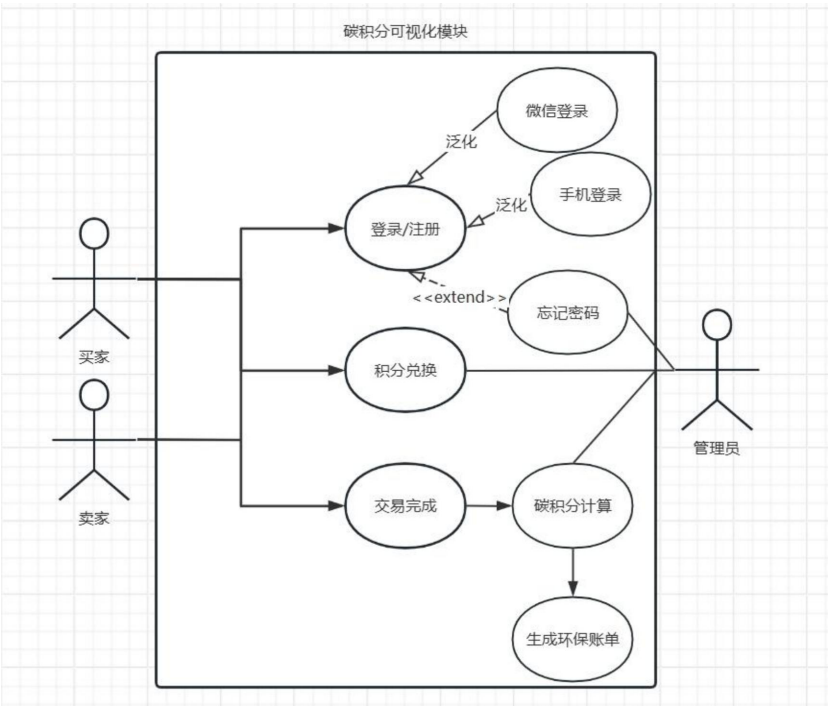


Figure 14 Use Case Diagram of Carbon Credit Visualization Module

(2) Transaction Function

In the use case diagram of the trading module, the system covers three types of participants: sellers, buyers, and administrators. As the starting point of the transaction, the seller submits book content through the "Upload Book Information" use case, triggering the "AI Review Book Quality" use case for information review. After the review is approved, the "Publish Book Information" use case is executed to display the book on the platform. After the buyer and seller reach a transaction intention, they complete the transaction through the "Conduct Transaction" use case, which derives two specific payment methods, "WeChat Payment" and "Alipay Payment", through generalization relationships to adapt to diverse transaction needs. After the payment in the transaction process is successful, the system triggers the "Generate QR Code" use case. After the buyer and seller conduct the transaction offline on their own, they scan the QR code, ultimately achieving "Transaction Completion". In addition, as the system management role, the administrator runs through the entire trading module to ensure that the transaction process is standardized and orderly, and to guarantee the stable operation of each use case link. This diagram fully presents the entire process from book upload and review,

transaction payment to ensuring transaction success, clearly constructing the functional interaction logic of the trading module.

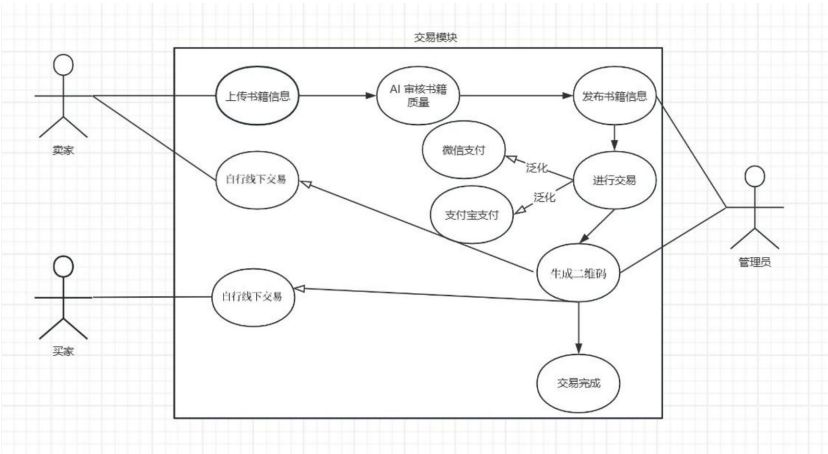


Figure 15 Use Case Diagram of QR Code Transaction Module

(3) Browse and search for classified books function

This is a use case diagram of the "Browse, Search, and Categorize Module" in the campus second-hand book circulation system, presenting the interaction logic between buyers, sellers, and the module's functions. The module is bounded by a rectangular box and covers four core functions: "Homepage Browsing", "Book Search", "Categorical Filtering", and "Uploading Books by Type". As participants, buyers and sellers can view book displays through homepage browsing, use book search to precisely find their targets, and narrow down the scope through categorical filtering; sellers can also upload books by type, standardize the entry of books for sale, contribute to the system's construction of an orderly book library, and provide functional support for the efficient matching of second-hand book resources between buyers and sellers.

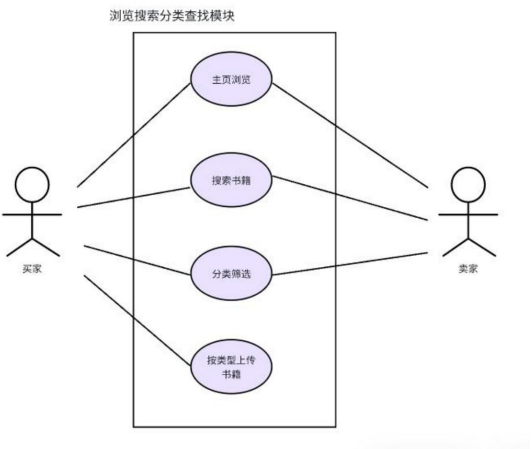


Figure 16 Use Case Diagram of QR Code Transaction Module

4.2.2 Functional Use Case Description

Through the campus second-hand book circulation system, users first log in and register, then, based on their own needs, utilize browsing, searching, and classification filtering functions to find eligible second-hand books, thereby resolving issues related to the acquisition or disposal of textbooks, academic materials, etc., covering scenarios such as freshmen seeking low-cost textbooks, graduates clearing books, and postgraduate students searching for scarce academic books.

Participant	Price-sensitive, graduation clearance, and academic research users
Precondition	The user has completed system registration and login, and the system has entered a certain amount Measure second-hand book information
Postcondition	1. Once the user finds the target book, they can enter the transaction process 2. If not found, you may adjust the search criteria or set a reminder. or participate in the release of purchase requests
Main Event Flow	
Browse Function	1. The user enters the system's home page, and the system follows the default classification (e.g., Textbooks, academic works, exam materials, etc.) or hot Door level, displaying a list of second-hand books, including book covers, Book title, price, brief description (version, condition, etc.), Seller information. 2. Users can scroll the page down to browse more books, click on a book entry, and enter the details page to view complete information.
	1. The user enters keywords in the search bar, and the keywords may include Book title, author, ISBN, course name, etc.

Search function	2. The system matches book information in real time and displays those containing keywords Second-hand book list, supporting sorting by relevance, price, and posting time Sorting, after the user selects the sorting method, view the corresponding search Search results, click to enter the details page.
Classification function	1. The system provides classification and filtering, including teaching materials and supplementary materials, as well as academic monographs et al.; 2. Users can also filter out second-hand books of the corresponding category Sort by price, condition, etc., browse the filtered results and View details.
Alternative event flow	1. No matching results 2. Book information is abnormal

Table 5 Use Case Description of Browsing and Searching Classification Function

From the seller uploading books and AI auditing the release information, to the buyer selecting the required books and then making a purchase, the entire process management of transaction execution (including online payment and offline transactions) and transaction completion covers various online and offline transaction scenarios, ensuring the efficient and standardized conduct of second-hand book transactions.

Participant	Buyers and Sellers
Precondition	The system has been connected to the AI review tool and has basic book information Storage and payment interfaces, etc., support QR code generation, and are fully Transaction rule configuration.
Postcondition	1. After the online transaction path is completed, the system records the complete transaction. Data, used for statistics and traceability; if offline transactions are retroactively recorded,

	Synchronize and update system data; otherwise, it is only a user-initiated action. The system has no mandatory records. 2. After the transaction is completed, the seller receives the corresponding payment, the buyer obtains the book, and the platform can carry out follow-up services based on the transaction data.
Main Event Flow	
Online transaction path	1. Upload Book Information: The seller triggers "Upload Book Information" Use case: Fill in book details and submit, then enter the system's pending review queue. 2. AI Review of Book Quality: The system automatically executes "AI Review Quality of nuclear books, verified based on image recognition and text rules Verify book information. If the review is approved, proceed to the next step; if not If it fails, provide feedback for the seller to revise / reject. 3. Publish Book Information: After passing the review, "Publish Book Information " use case triggered, book information listed and displayed for purchase Browse and search. 4. Conduct a transaction (linked after triggered by the buyer): The buyer selects a book and initiates a transaction, entering the "conducting a transaction" stage. The system provides two generalized branches, "WeChat Pay" and "Alipay", and the seller confirms the payment method and amount.
	5. Generate QR Code: The payment process is associated with "Generate QR Code

	code", used for offline settlement, and synchronously marks the transaction status as "To be completed" 6. Transaction Completion: The buyer confirms receipt of goods, triggering "Transaction Completion Successful "use case", the system records the transaction closed-loop and updates the order status state.
Offline transaction path for the buyer and seller	Offline self-transaction: Sellers go through online payment and AI review Nuclear and other processes trigger the "offline self-transaction" use case, After negotiating the transaction location with the buyer, complete the book settlement offline Answer.
Alternative event flow	1. AI review failed: If the book information violates regulations or is of poor quality Passed, AI review rejected, system prompts seller to modify content Resubmit, or directly terminate the publishing process, and the seller needs to Reorganize the book information and try uploading again. 2. Payment Exception: Select "WeChat Pay" or "Pay When using "Bao Payment", if network issues occur or account balance is insufficient Foot, payment interface failure, system prompts payment failure, leading Guide buyers to retry, change payment methods, or contact the administrator Troubleshoot problems. 3. Offline transaction disputes: After offline book handover, if there are issues between the buyer and seller Disputes over book quality and payment between the two parties require manual intervention for

	adjustment Solution, the system can provide transaction negotiation records to assist in judgment, shadow Confirmation of Transaction Completion Status
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Table 6 Use Case Description of QR Code Transaction Function

After the transaction is completed, the system automatically calculates the points for this transaction, generates environmental protection bills, and other functions, covering carbon credit management throughout the entire process of second-hand book transactions, contributing to environmental protection incentives and data visualization.

Participant	Buyers and Sellers
Precondition	The system has been associated with used book transaction data and supports retrieving transactions Completion status; has basic user information management, points rules Configures the function.
Postcondition	1. After successful login/registration, users gain system operation rights Limit; if the redemption is successful, the user's points will decrease and they will obtain the corresponding rights Benefit. 2. After the transaction is completed and carbon credits are calculated, the user's credit account Update, environmental bill data synchronization record, for subsequent visualization provides a basis for visualization and statistical analysis.
Main Event Flow	
	When a second-hand book transaction is completed (such as when the buyer confirms receipt or the platform determines the transaction is completed), the "Transaction Completed" use case is triggered. The system automatically

	executes "Carbon Credit Calculation" based on the preset rules
Transaction Completion and Carbon Credit Calculation	then (such as the carbon emission conversion formula for book circulation reduction), Calculate the carbon credits generated by this transaction for the buyer and seller, Synchronously update the user's points account.
Generate Environmental Protection Bill	After the "carbon credit calculation" is completed, the system triggers "generate Environmental Bill" use case, summarizing user data over a certain period (e.g., monthly) Details of carbon credit acquisition and utilization within a degree, quarter), generated into a visualized environmental protection bill, which users can log in to view and understand Personal environmental protection contribution data.
Points Redemption	After logging in, users can enter the "Points Redemption" function and Redeem rewards set by the platform based on existing carbon credits, which is System verifies the points balance and redemption rules, and completes the redemption deduction Integral.
Alternative event flow	Points redemption failed: insufficient points at the time of redemption, the inventory of the redeemed item is 0, or the rule verification fails (e.g., the time-limited redemption item has expired). The system will prompt the reason for the failure and guide the user to replenish points, change the item, or wait for the inventory to be updated.

Table 7 Use Case Description of Carbon Credit Visualization Function

The seller sends a video of the book, invokes the review rules through the AI inspection and review module for review, enters the transaction settlement module to issue instructions for

releasing the transaction funds, finally notifies the buyer of the review results, and the buyer finally updates the account status.

2. Carbon Credit Visualization Module

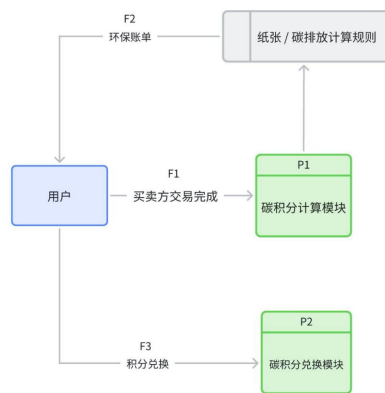


Figure 19 Use Case Diagram of Carbon Credit Visualization Module

The above figure is the data flow diagram of the carbon credit visualization module of our system. After the transaction between the buyer and seller is completed, an environmental protection bill will be generated. Environmental protection credits can be accumulated and redeemed for gifts in the credit mall.

3. QR Code Transaction Module

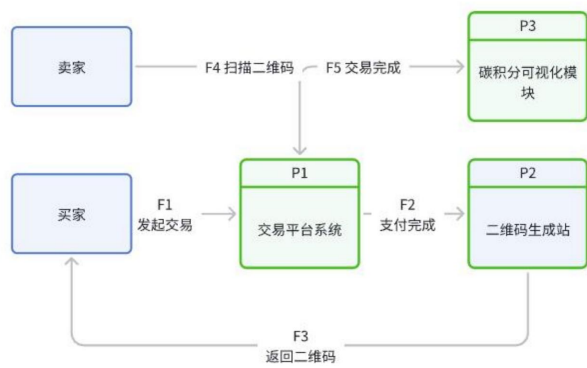


Figure 20 Use Case Diagram of QR Code Transaction Module

The trading platform system initiates a transaction, calls the QR code generation station to generate a QR code and returns it to the buyer; after the buyer completes the payment, the seller scans the QR code, the trading platform system marks the transaction as completed, and simultaneously triggers the carbon credit visualization module to realize the association between the second-hand book trading process and carbon credit calculation, completing all links from transaction initiation, QR code interaction to transaction closure and carbon credit linkage.

5 System Design

5.1 Overall System Design

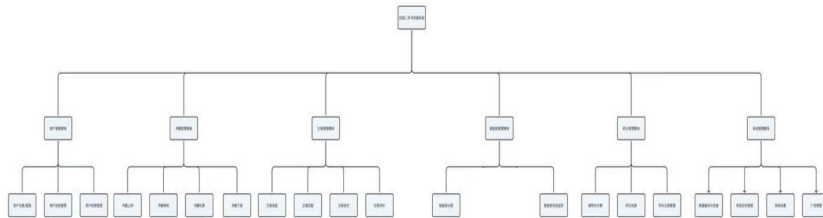


Figure 21 Overall System Design Diagram

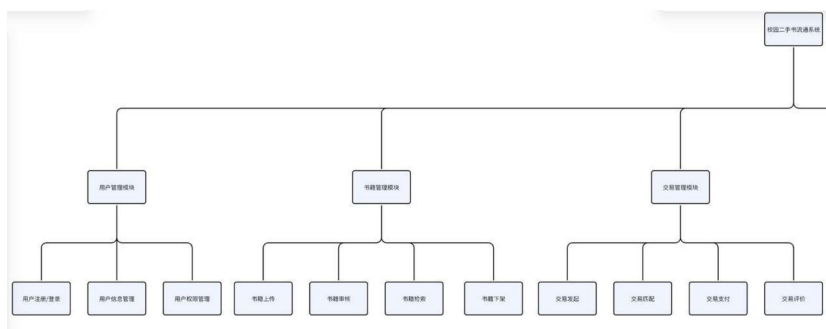


Figure 22 System Overall Design Subfigure 1

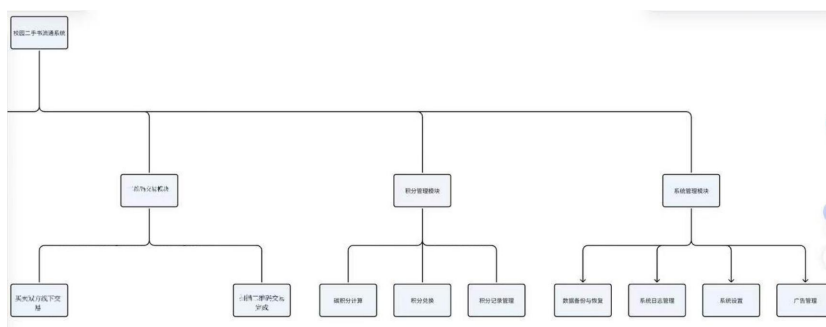


Figure 23 Subfigure 2 of the Overall System Design

The above is the structural diagram of our system, which divides the entire system into parts, with the function of each part being simple and clear. Using functional decomposition can simplify the system structure to a certain extent, making the system easy to understand and modify. The following is a textual description of the structural diagram:

1. User Management Module

User Registration / Login: Supports multiple login methods such as mobile phone number and WeChat, verifies information during registration, and encrypts and stores it.

User Information Management: Allows users to modify their personal profiles, and administrators can review and manage abnormal information.

User Permission Management: Distinguish roles such as ordinary users and administrators, and assign different operation permissions.

2. Book Management Module

Book Upload: Sellers upload book information, and the system automatically identifies the ISBN code to supplement details.

Book Review: AI combined with manual review of book information and copyright, marking abnormal books.

Book Search: Provides multi-dimensional search functionality, supporting precise and fuzzy queries for ISBN, book title, etc.

Book Removal: Sellers or administrators can remove books that violate regulations or are out of stock.

3. Transaction Management Module

Transaction Initiation: The buyer places an order, the seller confirms the order, and the system checks inventory and user information.

Transaction Matching: Match transactions according to user needs and preferences, and recommend relevant books.

Transaction Payment: Supports multiple payment methods, encrypts payment information, and ensures security.

Transaction Evaluation: After the transaction is completed, both the buyer and the seller evaluate each other, and the evaluation affects the credit rating.

4. QR Code Transaction Module

Offline transactions between the buyer and seller: When purchasing books, the buyer fills in the transaction location for subsequent offline transactions between the buyer and seller.

QR code scanning transaction completed: After the book handover is completed, the seller scans the QR code, and the successful transaction is automatically uploaded to the system.

5. Points Management Module

Carbon Credit Calculation: Calculate carbon credits based on transaction behavior, with adjustable calculation rules.

Points Redemption: Users redeem gifts or privileges with points, and administrators review redemption requests.

Points Record Management: Records changes in points, and users can query points details.

6. System Management Module

Data Backup and Recovery: Regularly back up data to ensure it can be restored in the event of a disaster, safeguarding data security.

System Log Management: Records system operation logs for troubleshooting and security auditing.

System Settings: Administrators configure system parameters, such as transaction fees and points rules.

Advertising Management: Manage advertising placements on the platform, review advertising content, and statistics advertising effects.

5.2 Class Design

5.2.1 Rough Design of Classes

1. QR Code Transaction Module

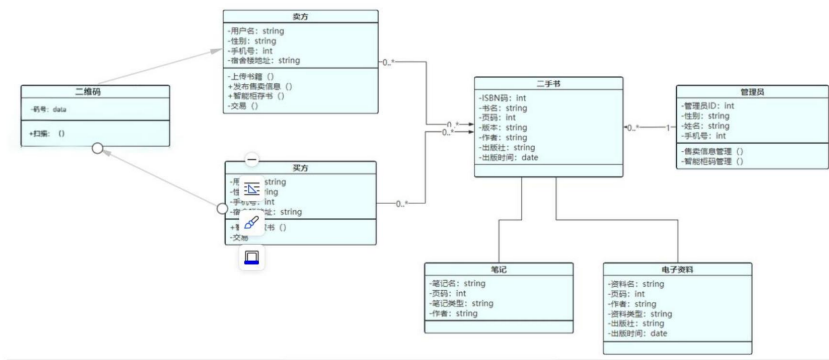


Figure 24 Rough Class Diagram of QR Code Transaction Module

This is the QR code transaction module class diagram of the campus second-hand book trading system, which is the core component of the system and covers User Personas, trading objects, and management modules.

Seller User Persona: It has attributes such as username, gender, mobile phone number, dormitory building address, etc. Through methods such as "uploading books", "publishing sales information", "storing books in smart cabinets", and "transactions", it is associated with smart cabinets, second-hand books, and buyers, and dominates the processes of book uploading, storage, and transaction initiation.

Buyer Class: Contains attributes such as username, gender, mobile phone number, dormitory building address, etc., depends on methods such as "Smart Cabinet Book Retrieval" and "Transaction"; interacts with smart cabinets and second-hand books to complete book retrieval and transaction operations.

Used Book Category: With ISBN code, book title, page number, edition, author, publisher, and publication date as core attributes, it serves as the main subject of transactions, associates

with sellers and buyers, and simultaneously establishes connections with Note Category and Electronic Material Category. The Used Book Category also includes Note Category: which attaches to used books through attributes such as note name, page number, note type, author, etc., to supplement transaction content; and Electronic Material Category: which associates with used books through attributes such as material name, page number, author, material type, publisher, publication date, etc., to expand the scope of transaction items.

QR Code Class: It has the attribute of a QR code number and can be scanned by the seller.

The entire core process class diagram defines business characteristics through attributes, implements specific functions through methods, constructs a complete interaction logic through association relationships, and provides architectural support for the entire process of the campus second-hand book trading system.

2. Carbon Credit Visualization Module



Figure 25 Rough Class Diagram of Carbon Credit Visualization Module

The following is an overview of the class diagram for the carbon credit visualization module of the campus second-hand circulation system

Seller: Attributes include username, gender, mobile phone number, dormitory building address. Methods are uploading books: entering book information (such as book title, ISBN, price). Publishing sales information: displaying book information on the platform for buyers to browse. Storing books in smart cabinets: depositing books into designated smart cabinets, requiring association of cabinet number and location. Transaction: completing operations such as order confirmation and payment collection.

Buyer: Attributes include username, gender, mobile phone number, dormitory building address. Methods include retrieving books from the smart locker: extracting books from the smart locker by scanning a QR code or using a verification code. Transaction: Completing operations such as placing an order, making a payment, and confirming receipt of goods.

Administrator: Attributes include administrator ID, gender, name, mobile phone number (same as user attributes).

Methods include Forgot Password operation: assist users in resetting passwords. Points redemption management: review points redemption requests and configure redemption rules. Carbon points calculation management: adjust points calculation rules and handle exceptions. Transaction management: monitor transaction processes and handle disputes or refunds.

Additionally, there is offline QR code scanning to ensure the completion of transactions. Transaction process design: The transaction method exists in both the seller and the buyer, but the specific implementation differs. Clarify the transaction process:

Buyer: Search for books → Place an order → Discuss payment location → Make payment → Generate QR code.

Seller: Confirm the order → Discuss the payment location → Receive payment → Scan the QR code → Transaction completed.

Points System Design: Administrators maintain the points system through points redemption management and carbon points calculation management.

5.2.2 Detailed Design of Classes

Based on the rough design of the above classes, we further analyzed the detailed design of the classes, which is divided into boundary classes, control classes, and entity classes.

1. Boundary Class

In the second-hand book trading system, users need to complete a series of operations in sequence: First, users can log in through WeChat or mobile phone, while new users can complete registration and enter the system by entering information; if users forget their passwords, they can reset new passwords using the password retrieval function to regain account access. Second, as the starting point of second-hand book trading, users upload book information and publish trading information. During the trading process, based on the trading QR code, the buyer can display the QR code for the seller to scan during offline interaction, ultimately completing the transaction. Additionally, the system supports users in conducting point redemption operations to exchange points earned from transactions for corresponding goods or services, thus forming a complete user operation process for the second-hand book trading system.

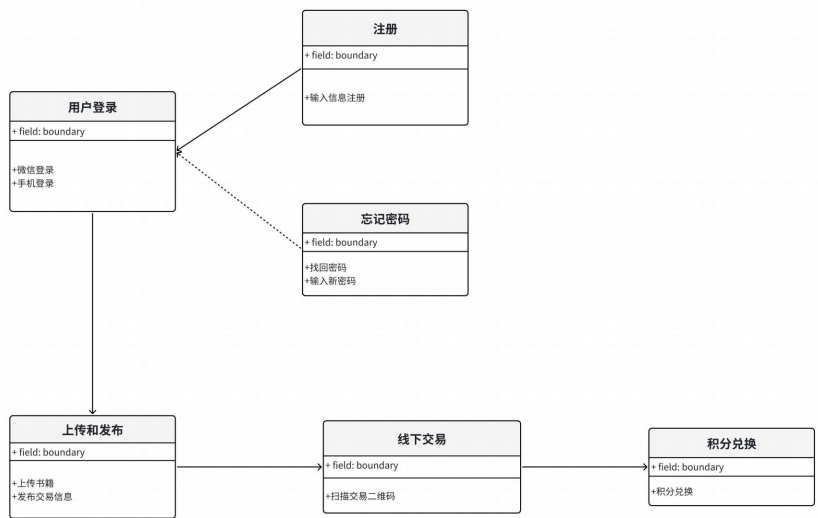


Figure 26 Boundary Class Diagram

2. Control Class

In the control process of the used book trading system, the process is first initiated by the AI book review stage, which relies on AI technology to check attributes such as the book's ISBN information, missing pages, note retention status, and degree of wear and tear.

Based on this, the basic condition of the book is accurately evaluated; then, it enters the transaction processing stage, where information on the entire process of second-hand book transactions is managed, including pre-publication review of transaction information, dynamic monitoring during the transaction process, and commission collection operations after the transaction is completed, ensuring that the transaction proceeds in a standardized and orderly manner; subsequently, the QR code generation stage is triggered, where the system automatically generates a unique code for the buyer and seller to complete the transaction, safeguarding the rights and interests of both parties; after the user completes the offline handover of the book, they scan the QR code, and the system confirms the final completion status of the transaction, while collecting the corresponding commission according to the rules, marking the phased conclusion of a single transaction process; based on the transaction completion data, the carbon credit calculation stage is initiated, where carbon credits generated by the transaction are quantitatively calculated based on key factors such as the number of pages of the book, contributing to the establishment of an environmentally friendly transaction incentive model; finally, the bill generation stage, based on the results of the credit calculation, accurately generates and publicly releases bills associated with the credits, providing a clear basis for subsequent credit settlement, query, etc., connecting the entire control process from book review to transaction closure and environmental incentive feedback.

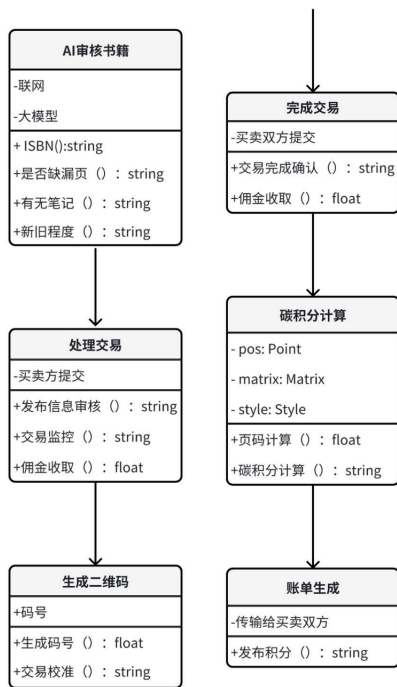


Figure 27 Figure 28 Control Class Diagram

3. Entity Class

In the entity architecture of the used book trading system, the core entity "used book" uses attributes such as ISBN, page number, publisher, author, and edition to carry the basic object information of the transaction and is associated with multiple entities. The "buyer", as the subject of the purchase behavior, has attributes such as name, gender, major, mobile phone number, and dormitory building address; the "seller", as the subject of the selling behavior, has the same attributes as the buyer, jointly forming the two parties involved in the transaction. The "note" entity uses entity, page number, and subject classification as attributes, serving as additional content for used books to enrich the value of the traded items; the "electronic material" entity contains the entity attribute and can be associated with used books to provide additional learning resources. The "QR code" entity uses the code number as a carrier to perform transaction confirmation and calibration; the "administrator" entity uses name, gender, ID, and mobile phone number as attributes and assumes the management responsibilities for QR codes and related business. The above entities jointly construct the entity model system of the used book trading system through attribute definitions and association relationships.

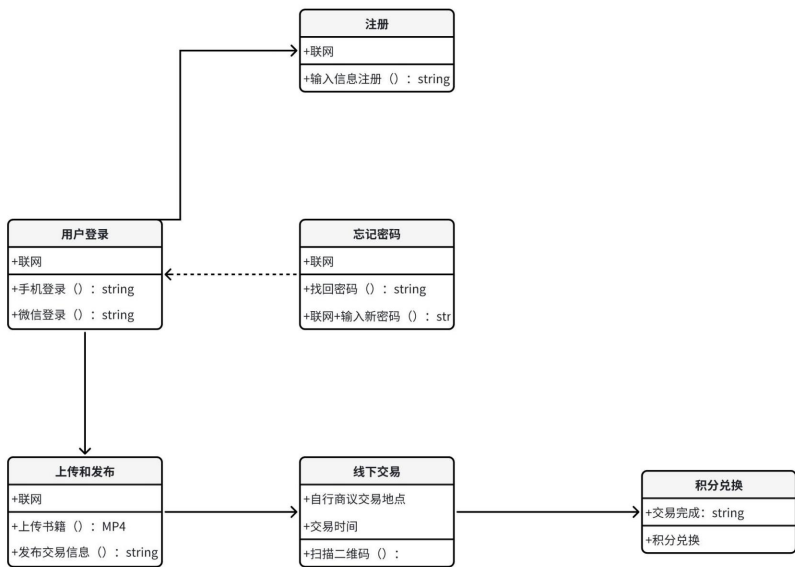


Figure 29 Entity Class Diagram

5.3 Sequential Design of System Core Modules

1. Transaction Module Sequence Design

In the sequence diagram of the second-hand book trading system, each role constructs a dynamic interaction system through vertical lifelines and horizontal message flows: the seller first uploads book information (including metadata such as ISBN and version) to the AI review system via asynchronous call

Data), the AI review system performs content verification based on the rule engine. After the review is approved, the AI review system sends an approval event to the trading platform. Sellers can then trigger the book information publishing process and write the product data into the distributed database of the trading platform.

On the buyer side, the user initiates a transaction request on the front end of the trading platform. The two parties negotiate the offline transaction location and time. After the buyer completes the payment, the trading platform triggers the QR code generation service through the event bus. After the two parties successfully complete the offline handover of the book, the seller scans the transaction QR code, and the transaction can be successfully completed. The record is automatically uploaded to the platform. Finally, the platform notifies both the buyer and the seller of the transaction completion through domain events and triggers the carbon credit calculation module to perform environmental incentive accounting.

This sequence diagram clearly presents the message transmission order and interaction logic among the seller, AI review system, trading platform, and buyer through vertical role lifelines and horizontal message arrows, which helps to understand the dynamic behavior of the trading module and the collaboration process of each link.

交易模块顺序图

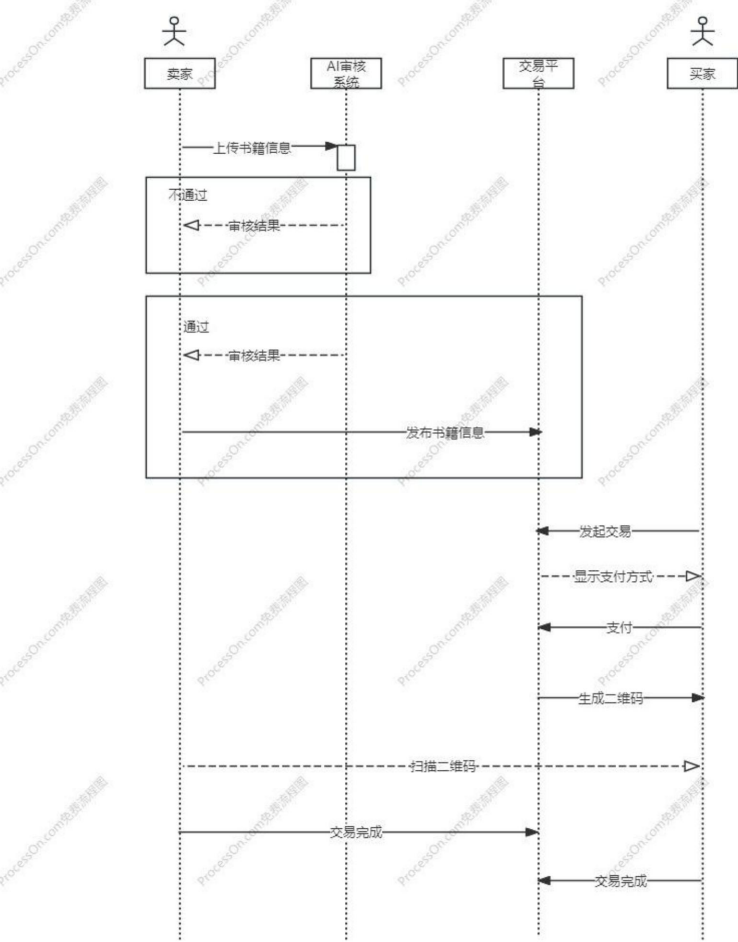


Figure 30 Transaction Module Sequence Diagram

2. Sequential Design of Carbon Credit Visualization Module

In the carbon credit management module of the used book trading system, the processes of transaction completion and credit redemption are implemented through an event-driven architecture: when both the buyer and seller complete the performance of their obligations on the trading platform, the trading platform sends a domain event containing transaction details to the carbon credit calculation module via a message queue. The module executes credit calculation based on a pre-set carbon emission conversion algorithm (e.g., each 100-page book circulation reduces 0.5kg CO2 equivalent), synchronizes the results to the user account system, and triggers

Environmental protection bill generation service. Bills are presented in record form, generating reports containing detailed information on points acquisition and redemption, and supporting users to view them on mobile devices.

In the points redemption scenario, users initiate a redemption request through the front-end interface, which is encapsulated by the points redemption module into an audit ticket containing points deduction rules and pushed to the administrator backend via the workflow engine. The backend executes the audit operation based on the rules engine, submits the audit

result (approved/rejected) through the form, and after the module receives the audit status, it ensures the consistency of points account deduction and prize distribution: when the audit is approved, it triggers the prize distribution process via the message middleware and pushes a redemption success notification to the user; if the audit is rejected, it returns the specific reason. This sequence diagram clearly presents the interaction order and logic of each role in the carbon points calculation and points redemption processes through vertical role lifelines and horizontal message arrows, which helps to understand the dynamic business process of the carbon points visualization module.

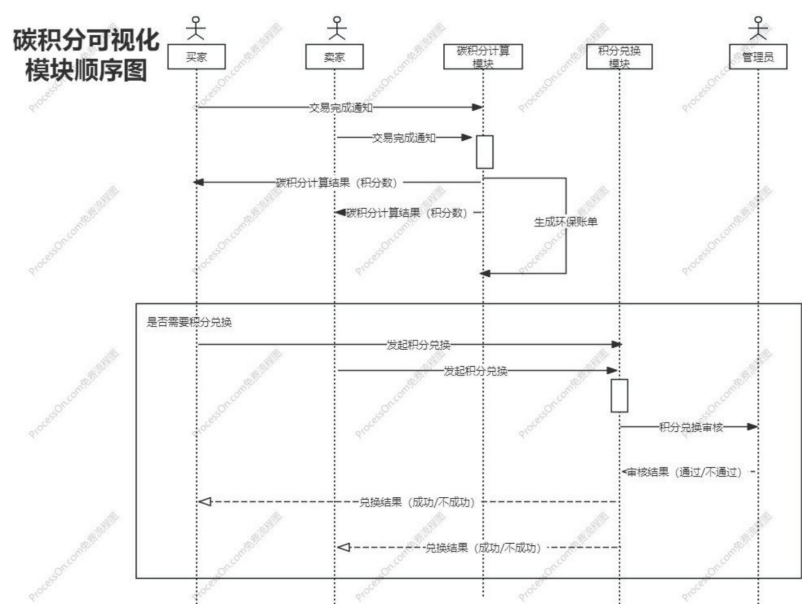


Figure 31 Carbon Credit Visualization Sequence Diagram

5.4 System Business Workflow Design

1. User Lifecycle Management Process

Registration and Identity Verification: New students / current students register via mobile phone number, enter campus card information (student ID, college), and complete real-name authentication registration.

2. Book Shelving and Review Process

① Seller Upload Process

Sellers use mobile devices to take photos and recognize ISBNs, automatically populate metadata such as book title, author, and publisher, supplement the condition (e.g., 90% new), note-taking status (including electronic materials with attachable files), and set the price (the system provides "textbook average price reference").

② AI Content Moderation Mechanism

Content Compliance: Identify whether the book title/content contains prohibited information (such as pirated or sensitive textbooks), and scan the cover/copyright page to verify the consistency of the ISBN and edition.

Quality Assessment: Image recognition detects book damage (missing pages, water stains), and in combination with user historical evaluation data, triggers manual review for high-risk sellers (e.g., if a seller uploads damaged books three consecutive times, they will enter the manual review pool).

Real-time feedback on review results: Push "Approved / Reason for Rejection" (e.g., "Version does not match description, please correct"), and rejected books automatically enter the draft box for modification.

3. Core Process of Transaction Closed Loop

① Search and Match

Buyers filter books by keywords (such as "Advanced Mathematics A Edition, 7th Edition") and categories, and the system recommends "hot-selling books in the same major" based on collaborative filtering.

Detail page display: actual photos of the book, seller's credit score, estimated carbon credits (e.g., "This transaction can earn 15 credits").

② Transaction Execution

After placing an order, the Payment Gateway is triggered, supporting "Escrow Transaction".

After the payment is successful, the trading platform generates a transaction QR code.

③ Closed-loop Offline Transaction

Supports the "face-to-face transaction" model: The buyer generates a QR code containing book information, and after the seller scans the code for confirmation, both parties complete the book handover offline, the transaction process is officially successful, and the system synchronously records the transaction data.

4. Carbon Credits and Environmental Incentive Process

① Automatic calculation of points

Upon completion of the transaction, the carbon credit module synchronizes to the user account via the event bus based on the book's page number (e.g., "This transaction earns 20 credits, with a cumulative total of 150 credits").

The system regularly generates an "environmental protection bill": it displays carbon emission reduction in the form of a form and supports sharing to social platforms.

② Points redemption system

Redemption process: User submits an application → Points redemption module verifies rules (such as points validity period, inventory) → Administrator reviews → Rewards are issued and points are deducted.

Through the above process design, the system can maximize the efficiency of second-hand book circulation, while integrating environmental incentives and campus contextualized experiences to form an ecological closed-loop of "upload - transaction - circulation".

5.4.1 Activity Design of System Core Modules

1. Activity Diagram of QR Code Transaction Module

After logging in through the system entry, the seller enters the book information upload interface, where they need to fully fill in the basic book information, including the book title, ISBN number, edition, condition, etc., and upload real photos of the book. After submission, the system activates the AI review mechanism to automatically verify the accuracy of the book information and the appearance condition of the book (such as damage, stains, etc.). If the review is approved, the book information will be automatically published on the trading platform for buyers to browse and select; if the review fails, the system will provide the seller with specific reasons, and the seller can modify the information according to the prompt and resubmit it for review. Books that fail the review will not be displayed publicly for the time being.

After browsing book information, the buyer selects the desired book and initiates a transaction. The system supports two mainstream online payment methods: WeChat Pay and Alipay. After completing the payment, the system automatically generates a QR code and pushes it to the buyer. After the buyer and seller complete the offline handover of the book, the seller scans the QR code, and the system confirms the transaction is completed.

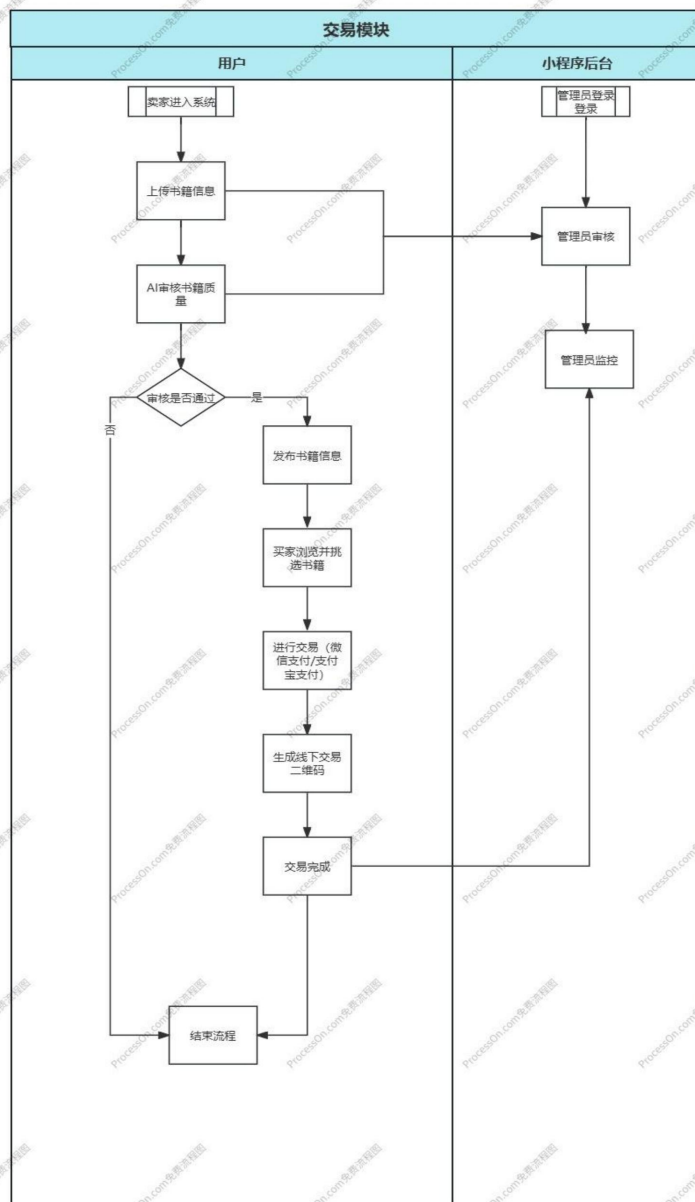


Figure 32 Activity Diagram of Transaction Module

2. Activity Diagram of Carbon Credit Visualization Module

After a user successfully logs in, the personal center includes the option to select redeemable goods or services and complete the process of deducting points and distributing rewards. Meanwhile, after a user completes a second-hand book transaction, the system will automatically trigger the carbon credit calculation function, calculate the carbon credit value generated by this transaction based on parameters such as the number of pages and circulation times of the traded book, and synchronize the credit data to the user's account.

After the carbon credit calculation is completed, the system will automatically generate an environmental protection bill, summarizing the user's carbon credit acquisition, usage details, and corresponding environmental protection contribution data (such as reduced carbon emission equivalents) within a certain period. After the bill is generated, users can view,

download, or share it through the mini-program interface, thus concluding the credit operation process. If the system determines that the user's identity does not belong to the buyer or seller role, it will directly terminate the credit-related operations and end the process.

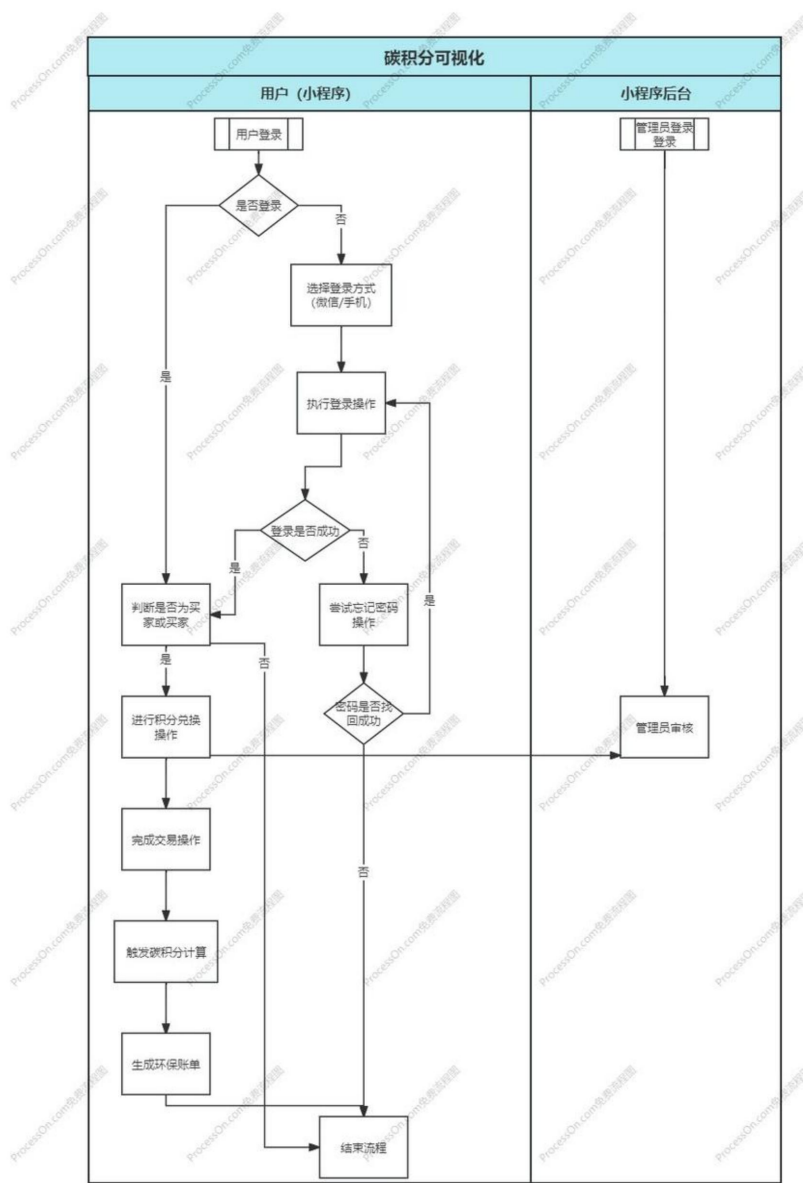


Figure 33 Carbon Credit Visualization Activity Diagram

5.5 LOGO Design

Our system, named "Campus Book Recycling", is a campus second-hand book circulation system. As the name suggests, it conducts campus second-hand book transactions within the school.

Its simple logo presented in black and white style, with the pattern part consisting of three stacked books forming the base, and a flying bird outlined by simple lines above, whose body is ingeniously integrated with the outline of the books, creating a visual isomorphism of "books

are flying birds, flying birds carry books"; below it is accompanied by the sans-serif English "Books and Circulation", which also emphasizes the core function of the system again, strengthening the visual tension through the strong contrast between black and white. The stacking of the three books forms a stable triangular structure, and the tilt angle of the bottom book gives the picture a sense of dynamics. The flying bird's head faces right with slightly spread wings, symbolizing that books, like birds delivering information, flow nimbly among different users. The black and white color scheme not only highlights the classic and professional visual temperament but also ensures high recognition on the second-hand book trading platform through minimalist lines, accurately conveying the core concept of "book circulation".



Books and Circulation

Figure 34 LOGO Diagram

5.6 Page Design

5.6.1 Login Page

This is the login interface of the "Campus Book Recycling" platform, with its overall design closely centered on the theme of eco-friendly second-hand book trading and presented in a simple and fresh style. The interface background uses a soft light green gradient, conveying the concept of green and environmental protection, which echoes the platform's slogan "Let knowledge circulate and make the earth greener", creating a comfortable and natural visual atmosphere.

At the top center position, the platform name "Campus Book Recycling" is displayed, paired with a simple circular arrow icon to strengthen the brand identity; the slogan below, "Let knowledge circulate, make the Earth greener," further elaborates on the platform's core values.

The middle area is the login function module, highlighted in the form of a white card to create visual focus. Inside the card, the title "Welcome Back" is kind and friendly, guiding users to log in to their accounts and start their second-hand book trading journey. In terms of login methods, two options, "Login with Mobile" and "Login with WeChat", are provided to meet the

usage habits of different users; after choosing to log in with mobile, it is further divided into "Login with Password" and "Login with Verification Code" to adapt to diverse needs. The input area clearly demarcates the "Mobile Number" and "Password/Verification Code" fields, with an eye icon next to the password field to support password visibility toggling, enhancing interaction convenience; the "Forgot Password" link provides a retrieval path for users who have forgotten their account passwords, improving the account management process.

At the bottom, there is a guiding text of "Sign Up Now", with a clear entry point to help new users complete the registration conversion; meanwhile, it displays "Logging in means agreeing to our Terms of Service and Privacy Policy" in smaller font, complying with regulatory requirements and safeguarding user rights.

The overall interface layout is well-structured, with harmonious color coordination. The functional design aligns closely with user usage scenarios, simply and intuitively focusing on the core login process, enabling users to quickly get started and initiate the second-hand book trading experience, thereby conveying the platform's professional, environmentally friendly, and user-friendly brand image.

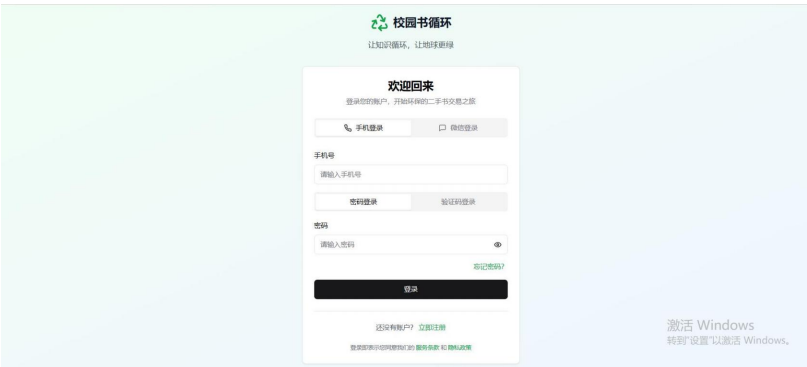


Figure 35 Login Page Diagram

5.6.2 Registration Page

The registration process of the "Campus Book Recycling" platform consists of three steps, with the interface featuring a fresh light green background to create an environmentally friendly and simple atmosphere. The first step, "Basic Information," uses a white card to hold required fields such as name and mobile phone number, including a password visibility toggle and mobile verification code sending function. Re-entering the password reduces the risk of setting an incorrect password due to input errors.

The "Next" button at the bottom drives the process forward; Step 2, "Profile", continues the card layout, requiring users to fill in campus attribute information such as student ID and major to help the platform accurately associate with campus scenarios, with "Next" leading to the final step; Step 3, "Complete Registration", displays a summary of the filled information on the card, highlighting new user benefits, and after checking agreements such as the Terms of Service and

Privacy Policy, clicking "Complete Registration" will conclude the process. The overall layout is clearly structured, with the registration stages intuitively presented through a Progress Bar, and the color and functional design align with the environmental and convenient positioning of the campus second-hand book trading platform.



Figure 36 Registration Page Figure 1



Figure 37 Registration Page Figure 2

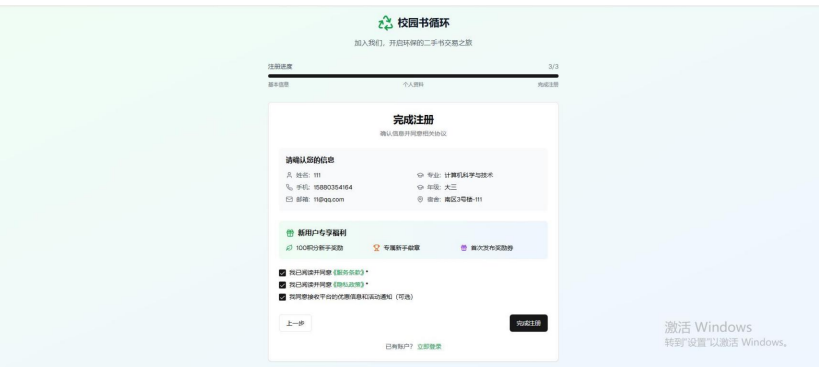


Figure 38 Registration Page Figure 3

5.6.3 Home Page

The homepage of the "Campus Book Recycling" platform is designed to be simple and intuitive, creating an environmentally friendly atmosphere with fresh color tones. The top navigation bar has clear functions, with options such as "Homepage, Publish Books, QR Code Transaction, Carbon Credits" clearly arranged, and convenient access to the login and personal center. The gradient in the middle conveys the concept of "knowledge recycling, green earth",

and the search bar supports searches by book title, author, and ISBN, quickly leading users to the target books. The data module below displays information on books for sale, registered users, successful transactions, and carbon reduction, intuitively presenting the platform's ecosystem; the popular categories and featured books sections, the former guides users by categorizing books into textbooks, postgraduate entrance exam materials, etc., enabling quick access to categorized books for easy retrieval. The latter recommends high-quality second-hand books in a combination of text and images, accompanied by a "View Details" button that redirects users to the book details page. From visual layout to functional design, it helps users efficiently conduct second-hand book transactions, practicing the dual values of environmental protection and cost savings.



Figure 39 Home Page Figure 1

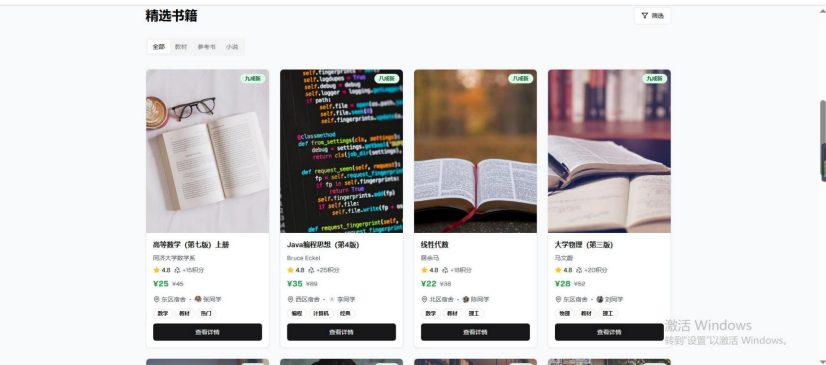


Figure 40 Home Page Figure 2

5.6.4 Release Book Page

The "Publish Used Books" page focuses on the demand for efficient and standardized circulation of idle books, adopting a minimalist and clear layout style. The top title "Publish Used Books and Revitalize the Value of Idle Books" highlights the functional value, and the Progress Bar below breaks down the publishing process into five major steps - "Basic Information, Upload Images, AI Review, Publishing Confirmation, and Publishing Success" - in an intuitive segmented format, allowing users to clearly perceive the operation progress and remaining steps.

In terms of functional design, each step is closely linked and rich in details: In Step 1, the ISBN scanning function can quickly retrieve basic information through the book's code, significantly reducing the workload of manual entry. If the code recognition is abnormal, it also supports manually filling in fields such as book title, author, publisher, and edition, and can supplement details about the book's condition (e.g., 90% new), selling price, and description (marking defects such as notes and missing pages), enabling buyers to accurately understand the book's status.



Figure 41 Publishing Book Page 1

Step 2: Open the image upload entry, encourage users to submit photos from multiple angles such as cover, inner pages, and defective areas to endorse the authenticity of the transaction, and at the same time provide a considerate reminder that "clear photos help improve the transaction success rate" to guide users to optimize the content they post.



Figure 42 Publishing Book Page 2

Step 3 introduces AI Content Moderation, which automatically verifies information from dimensions such as information completeness (e.g., whether all required fields are filled), image quality (whether it is clear and distinguishable), and copyright compliance. After passing, it will also display the audit results (e.g., accurate information, clear images, etc.) and the estimated carbon credits, quantitatively presenting the environmental value.



Figure 43 Publishing Book Page 3

Step 4 provides a "Publish Preview" that simulates the display style of the book on the platform, including price, information, points, etc., allowing users to double-check key content one last time. After confirming that everything is correct, click "Confirm Publish" to proceed.



Figure 43 Publishing Book Page 3

Upon reaching the "Publication Successful" page, it not only displays core records such as book ID, publication time, status (On Sale), and points rewards, but also provides a book link for easy sharing, paired with buttons like "View Book Details", "Manage My Books", and "Continue Publishing", thus forming a complete functional closed loop from publication to management.

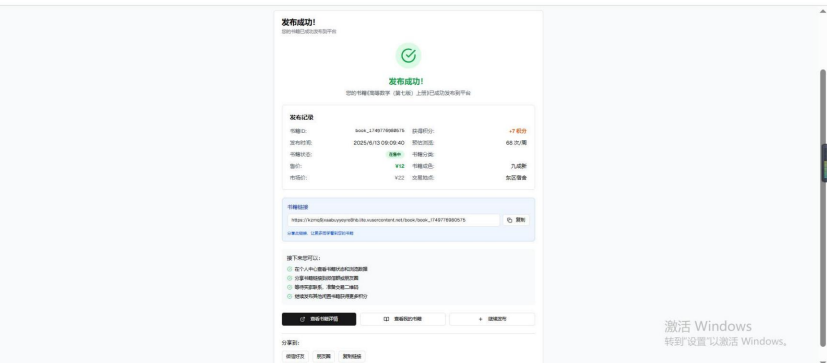


Figure 44 Publishing Book Page 4

The overall design centers around User Experience, using clear process guidance, intelligent information assistance, and a complete functional closed-loop to lower the threshold for second-hand book posting. This not only ensures the standardization and transparency of transaction information, helping users efficiently revitalize their idle books, but also deepens

the platform's concept of "knowledge circulation, environmental protection" through designs such as carbon credits.

5.6.5 Book Purchase Page

The book purchase page is designed around the second-hand book trading scenario, with a clear and intuitive layout.

Taking the details page of *Thinking in Java (4th Edition)* as an example, the left side displays multi-dimensional images of the book (cover, back cover, inner pages, etc.) and descriptions, the middle presents basic information and descriptions such as the book title, author, price, etc., the right side aggregates seller information (credit rating, transaction data), transaction guarantees (AI review, after-sales rules), and operation entry points, and there are also related recommended books at the bottom, which can recommend similar books to users, making book purchasing convenient and fast.

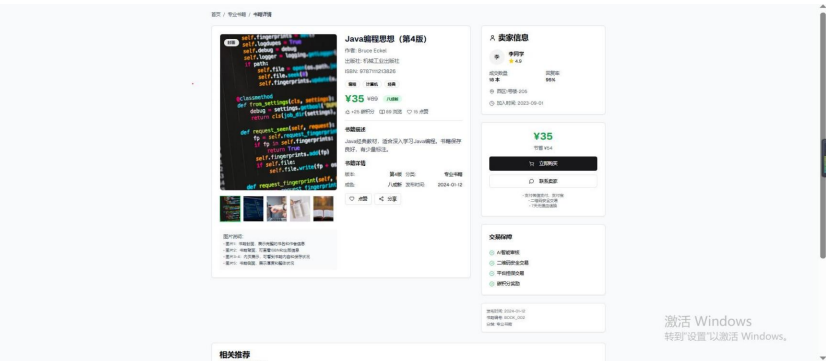


Figure 45 Purchase Books Page 1

Functionally, clicking "Buy Now" pops up an order placement window, supporting customizing the transaction location and time, filling in a message, and selecting WeChat/Alipay payment; after submission, it enters the order confirmation page, displaying order information and generating a payment QR code. Upon successful payment, it provides feedback on the transaction order number, carbon credit rewards, etc., and also allows viewing and using the transaction QR code for offline settlement.



Figure 46 Purchase Books Page 2

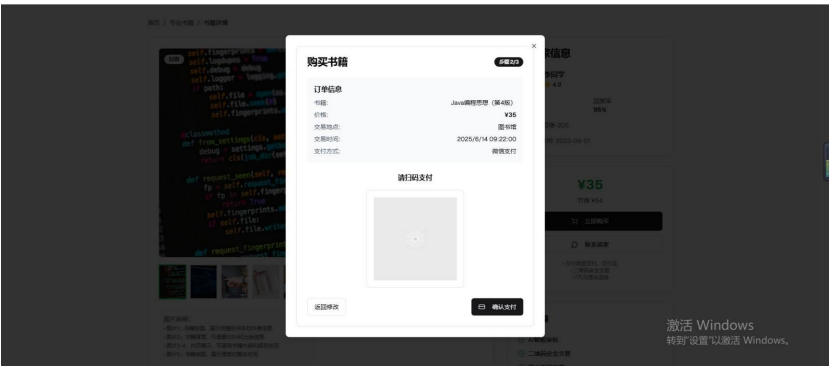


Figure 47 Purchase Books Page 3



Figure 48 Purchase Books Page 4

After successful payment, the transaction status is synchronously updated to "booked" to prevent other users from making a second purchase, which could cause unnecessary trouble, and the seller's contact information is displayed, forming a complete closed loop of "browsing - placing an order - payment - settlement". This not only eliminates the buyer's concerns through detailed information display but also ensures the convenience and security of transactions with streamlined functions, meeting the actual needs of campus second-hand book transactions.



Figure 49 Purchase Books Page 5

5.6.6 QR Code Transaction Page

The QR code transaction page, with its concise and clear layout, focuses on the order management and settlement process of second-hand book transactions. At the top of the page, the title "Transaction Records" is displayed, and below it, transaction information is listed by order, including basic details such as book title, price, seller, transaction location, etc. Each order is associated with a dedicated transaction QR code, accompanied by "Copy" and "Download" functions for easy retention and use.

In terms of functional design, after the buyer completes the payment, they can use the displayed QR code for offline delivery with the seller, and scanning the QR code will automatically complete the delivery. After the delivery is completed, there is also a "Confirm Delivery/Receipt" button, which is convenient for situations where the QR code has expired or there is a scanning error. Marking the order status, if there are issues with the transaction, users can also "Contact Buyer/Seller" to communicate, forming a complete closed loop from order viewing, QR code usage, status confirmation, to after-sales communication, ensuring the convenience and standardization of the offline delivery process in campus second-hand book transactions with a lightweight layout and practical functions.



Figure 50 QR Code Transaction Page 1

The QR code transaction module is also equipped with QR code scanning verification and detailed instruction guides to explain the transaction process to users. Under the "User Guide" tab, the operation processes for sellers and buyers are presented separately, with step-by-step instructions clearly showing all aspects of the QR code transaction; at the same time, safety tips (such as choosing a secure transaction location, verifying payment receipt, etc.) and frequently asked questions (QR code expiration, abnormal QR code scanning handling) are supplemented. Through text-based and process-oriented functional design, it helps users quickly understand and use QR code transactions in a standardized manner, resolve operational doubts, and ensure smooth campus second-hand book transactions.



Figure 51 QR Code Transaction Page 2

5.6.7 Carbon Credits Page

The carbon credit visualization page uses a fresh light green as the background color to create an environmentally friendly atmosphere. At the top, colorful cards intuitively display core data such as total credits, credits this month, ranking, and carbon reduction; the "Current Level" module below shows

User level progress and the points required for upgrading motivate users to accumulate points. Switching to the "Points Record" tab allows users to view details of points earned from completed transactions, first-time postings, etc.; switching to the "Leaderboard" presents the points rankings of environmental protection experts, strengthening competition and interaction.

Functionally, through data visualization and multi-dimensional display, users can clearly understand their carbon credit status. By means of level progression and ranking comparison, users are motivated to participate in second-hand book transactions, practice environmental protection actions, and achieve the value closed-loop of "knowledge circulation + green carbon reduction".

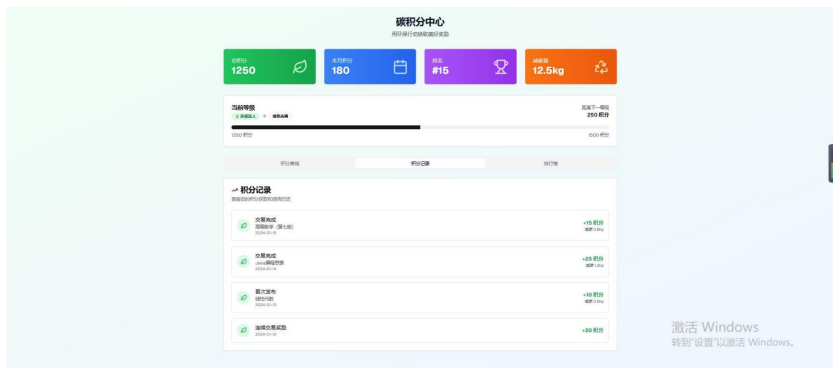


Figure 52 Carbon Credits Page 1

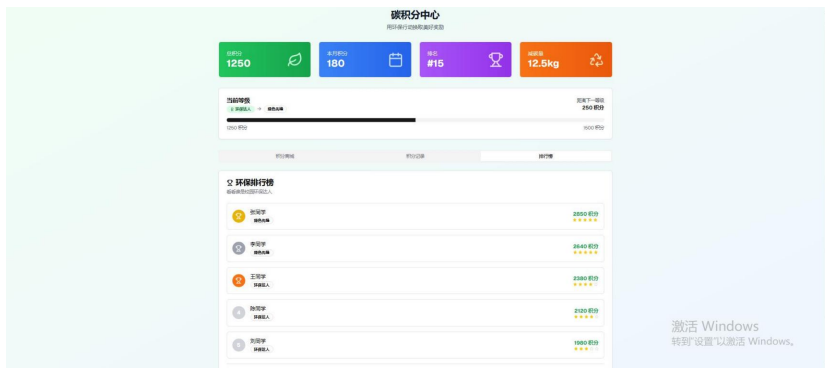


Figure 53 Carbon Credits Page 2

5.6.8 Points Redemption Page

The redemption page continues the platform's fresh style, with the core carbon credit data retained and displayed at the top. Below, the "Points Mall" section presents redeemable benefits in the form of category tags + product cards, such as Starbucks coffee vouchers, bookstore vouchers, etc., clearly indicating the credit consumption and inventory.

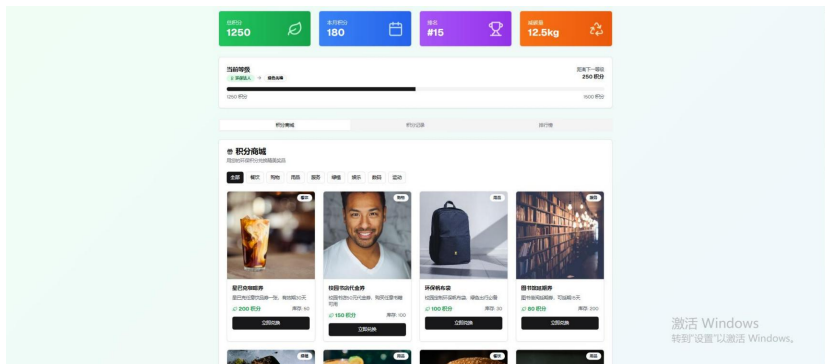


Figure 54 Points Redemption Page 1

Functionally, clicking "Redeem Now" on a product pops up a confirmation window, displaying product details, points changes, and collection methods. For dormitory delivery, personal information must be filled in. The rule that redemption cannot be revoked after confirmation is clearly stated; after confirmation, it redirects to the "Redemption Successful" page, generating a redemption code, usage instructions, and order records, supporting copy and sharing, enabling users to redeem practical campus scene benefits with carbon credits. Through the closed loop of "accumulation - redemption - feedback", users are encouraged to continuously participate in the second-hand book cycle, strengthening the platform's environmental protection value.

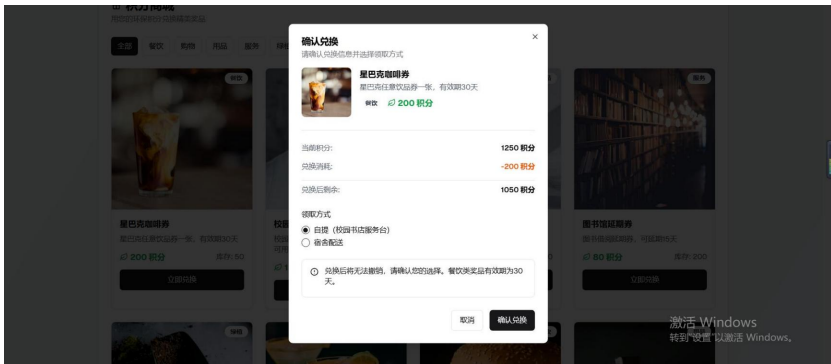


Figure 55 Points Redemption Page 2

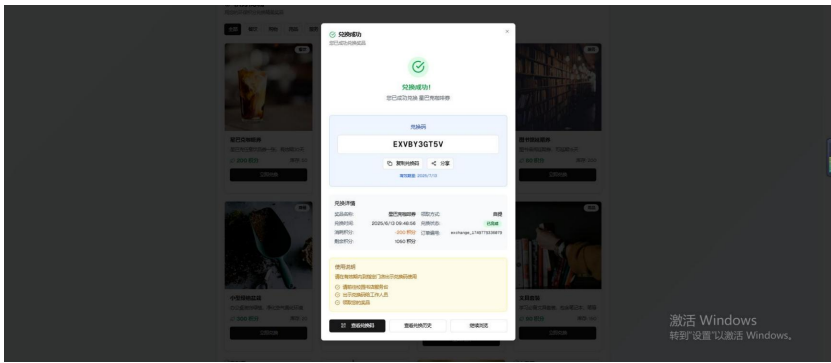


Figure 56 Points Redemption Page 3

5.6.9 Personal Center Page

The Personal Center page integrates users' core information and operations with a concise and clear layout. On the left side, basic personal information is displayed, while the "My Statistics" section below presents total points, transaction count, and credit score, intuitively showing users' participation results on the platform; on the right side, there are tabs for "My Books", "Transaction Records", and "Account Settings". Under "My Books", users can view the on-sale or sold status, price, and browsing and like data of published books, and can also click "Publish New Book" to continue transactions. "Transaction Records" and "Account Settings" cover the needs for transaction review and account management.

Functionally, through the information aggregation and classification operation entry, users can conveniently manage the entire process of second-hand book transactions, from posting to after-sales, from data viewing to account maintenance, forming a personalized and one-stop personal transaction management space that conforms to the user operation habits of campus second-hand book transactions.

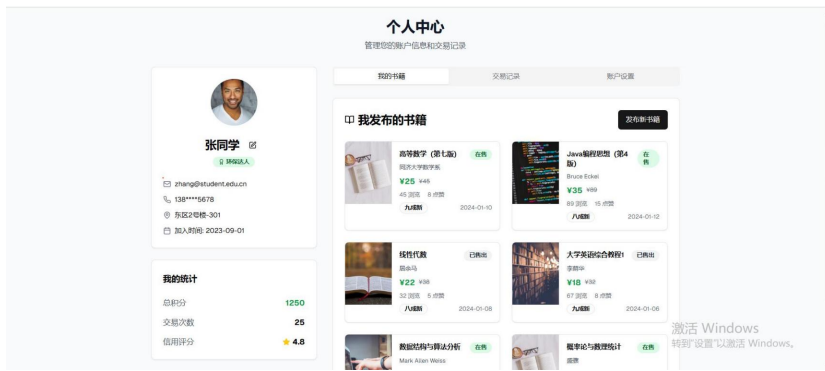


Figure 57 Personal Page

5.6.10 System Link

The campus second-hand book circulation system also comes with screen size Self-Adaptation. Click the following link, copy it to your browser, and it can be used on both PC and mobile devices:

<https://kzmq9jxaabuyyeyre8hb.lite.vusercontent.net/>

6 System Implementation

6.1 Introduction to Development Tools

Flowith, Doubao, V0.dev

Flowith is an AI productivity tool designed to enhance users' efficiency in content creation and help them more easily enter the flow state. Through its innovative node-based interaction method, combined with advanced AI technology, it provides users with an intuitive and efficient platform for knowledge exploration and learning. Users can freely create and connect nodes on an infinite canvas, with each node representing an independent question or topic, enabling the leap and divergence of thinking. It has a built-in large language model that integrates internet information through AI search to assist users in in-depth reading and exploration. It also has a built-in advanced AI image generation technology that can generate high-quality images based on user descriptions. It breaks the limitations of traditional search tools and provides a more free and creative way of information exploration. By combining multiple advanced AI models, it provides comprehensive and accurate information to help users build a complete knowledge system. Users can install different plugins according to their needs to expand the tool's functionality. The infinite canvas and node-based interaction provide users with an immersive experience, maintaining the coherence and concentration of thinking. It supports multiple AI models, with some models provided for free, reducing users' usage costs.

Doubao is an intelligent assistant software based on artificial intelligence technology, developed on the basis of ByteDance's Yunque model, with multiple functions such as chatbot,

writing assistant, English learning assistant, etc. It can quickly and accurately search for and provide relevant knowledge and information. Users only need to enter a question or keyword in the search box, and Doubao can generate relevant answers or information. It can help users complete writing in various styles, including primary school compositions, teaching designs, etc. Users only need to input writing requirements, and Doubao can generate text that meets the requirements. It has powerful computing capabilities, can solve various mathematical problems, and provide detailed problem-solving methods and ideas. For researchers and students, it can quickly find relevant academic literature, papers and other resources, providing comprehensive academic information. It supports mutual translation of multiple languages, including Chinese, English, French, etc., helping users break through language barriers.

V0.dev is based on natural language processing and deep learning algorithms. When users enter descriptive text prompts in the dialog box, the AI engine will parse the input, understand the intent, and generate React components that meet the requirements according to preset templates and rules.

Code. The code has good readability and maintainability. When processing images, it utilizes deep learning models such as convolutional neural networks to preprocess and parse uploaded images, identify UI elements, and extract layout information and style features. Users can customize or modify the generated UI through natural language descriptions, and NLP technology converts these descriptions into computer-understandable instructions to guide code adjustments. Based on user text prompts, React code that meets the requirements is automatically generated, allowing developers to easily make subsequent modifications and optimizations. The generated code undergoes optimization processing, which helps reduce potential bugs and performance issues and improves the overall quality of the project. It supports image processing, enabling users to improve the design of selected elements based on image content, facilitating intuitive understanding of the code and further optimization and adjustment. It provides a real-time preview function, allowing users to view the generated UI interface in real-time when inputting text prompts, promptly identify and correct issues, and improve development efficiency. It has a history record function for user-generated interfaces, making it convenient for users to view previously generated code, conduct comparisons and analyses, better understand the working principle of the AI engine, and optimize and adjust accordingly.

6.2 System Equipment Environment

Operating System: Win11;

Development tools: Flowith, Doubao, V0.dev;

6.3 System Development

6.3.1 Code Process Design

1. Login/Register

(1) Start: The starting point of the process.

Select Login Method: Users need to decide here whether to register or log in.

(2) Registration Process

Enter account/password/phone number: If you choose to register, you need to enter relevant information.

Verify uniqueness: Query the database to check if the entered account has already been registered.

is: If already registered, prompt "Duplicate Prompt" and return to restart.

No: If not registered, perform encrypted storage, encrypt the password using methods such as MD5, store it in the database, and then redirect to the login page.

(3) Login Process

After selecting to log in, you will directly enter the identity verification process, which is verified through methods such as account password verification or WeChat authorization.

Success: If the verification is successful, generate a Token, then jump to the home page, and the process ends.

Failure: If verification fails, prompt "Error Prompt" and return to restart.

(4) End: The end point of the process.

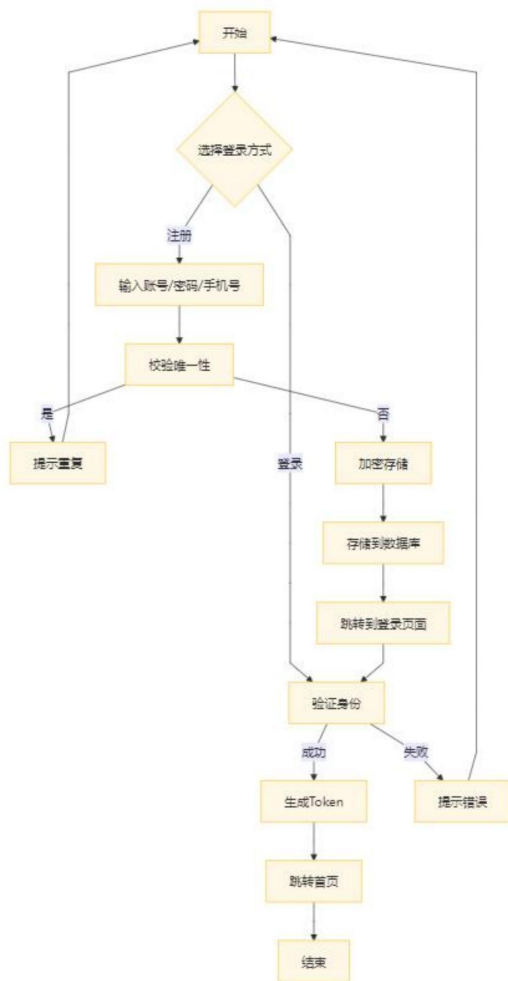


Figure 58 Login/Registration Code Flowchart

2. Book Upload and Review

(1) Upload Phase: The seller initiates the process, submitting basic book information (book title / author / price, etc.) and image materials (cover / copyright page).

(2) Content Moderation: The system automatically identifies the ISBN code through OCR technology and verifies the genuine information of the book:

Passed: Proceed directly to the publishing stage, and book information will be displayed on the platform.

Failed: Triggered manual intervention, with administrators conducting in-depth verification.

(3) Manual Review: For books with doubts in AI Content Moderation, manual verification of content completeness, copyright legality, etc. will be conducted: Pass: Complete publication; Fail: Reject to the seller, requiring modification and resubmission (the process reverts to the starting point).

(4) Closed-loop design: Books that are rejected for revision must undergo the full review process again to ensure that the book information on the platform is compliant and accurate.

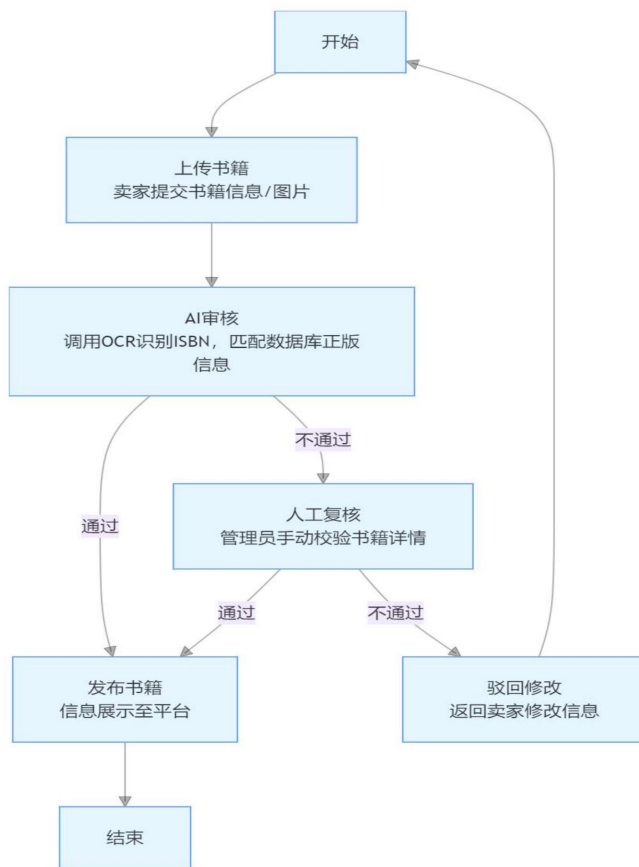


Figure 59 Flowchart of Book Upload and Review

3. Transaction Process

(1) Place an order and pay: The buyer selects books, confirms the order, enters the payment process, which supports both WeChat and Alipay payment methods, and completes the payment through a third-party payment API.

(2) Intelligent Allocation: After successful payment, the system automatically generates a transaction QR code, and both parties conduct offline transactions.

(3) Seller and Buyer: The two parties conduct transactions by scanning each other's QR codes or entering the transaction code offline.

(4) Carbon credit closed-loop: After the user confirms receipt of the goods, the system invokes the carbon credit calculation rules (such as green logistics carbon reduction credits) based on the order amount/behavior to complete the process closed-loop.

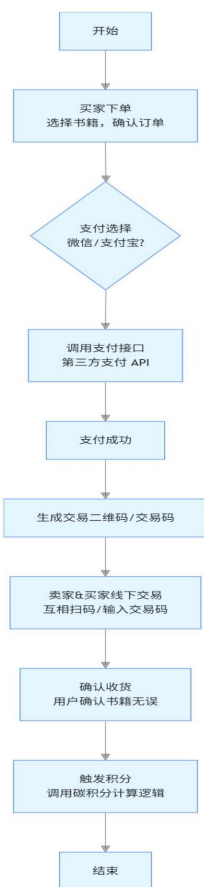


Figure 60 Transaction Code Flowchart

4. Carbon Credit Calculation Process

(1) Trigger condition: When the buyer and seller confirm the completion of the transaction (e.g., the buyer picks up the book and confirms receipt), the process automatically starts.

(2) Points calculation: The system applies a preset points formula (e.g., $\text{number of pages} \times 0.1 + \text{price} \times 0.5 = \text{carbon points}$) based on parameters such as the number of pages and price of the book.

(3) Bill Generation: Generate an environmental protection bill that includes details of points increase and decrease (e.g., displaying "50 points earned this time, cumulative points 1200"), which can be used for user inquiries or platform statistics.

(4) Account Synchronization: Real-time update of the user's points account balance, with points available for subsequent consumption deductions or gift redemption.

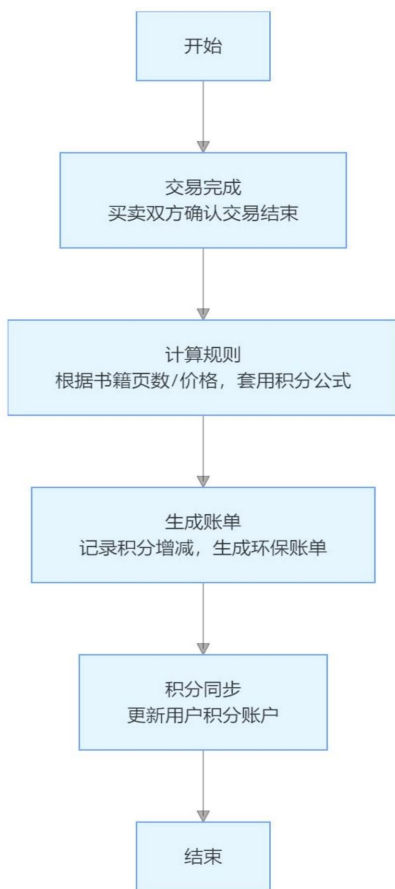


Figure 61 Flowchart of Carbon Credit Calculation Code

5. Points Redemption Process

(1) User-triggered: The user actively queries points and enters the gift redemption page to browse the list of redeemable gifts (such as books, cultural and creative products, etc.).

(2) Inventory Verification: The system verifies the inventory of the selected gifts in real-time:

Sufficient: Deduct the corresponding points from the user, and simultaneously update the gift inventory quantity (e.g., "remaining inventory - 1");

Deficiency: Prompt the user that "the gift is out of stock" and return to the redemption page to reselect.

(3) Result Notification: After the redemption is completed, the user will be informed of the success/failure result of the redemption via SMS or inbox notification, closing the process loop.

(4) Exception handling: When out of stock, it supports users to reselect gifts to avoid interrupting the operation process.

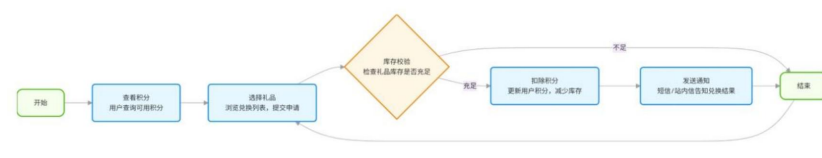


Figure 62 Flowchart of Points Redemption Code

6. Offline QR Code Transaction Process

(1) Start: The starting point of the process, choose to deposit or retrieve books

(2) Book Storage Branch:

The seller obtains the transaction QR code after payment

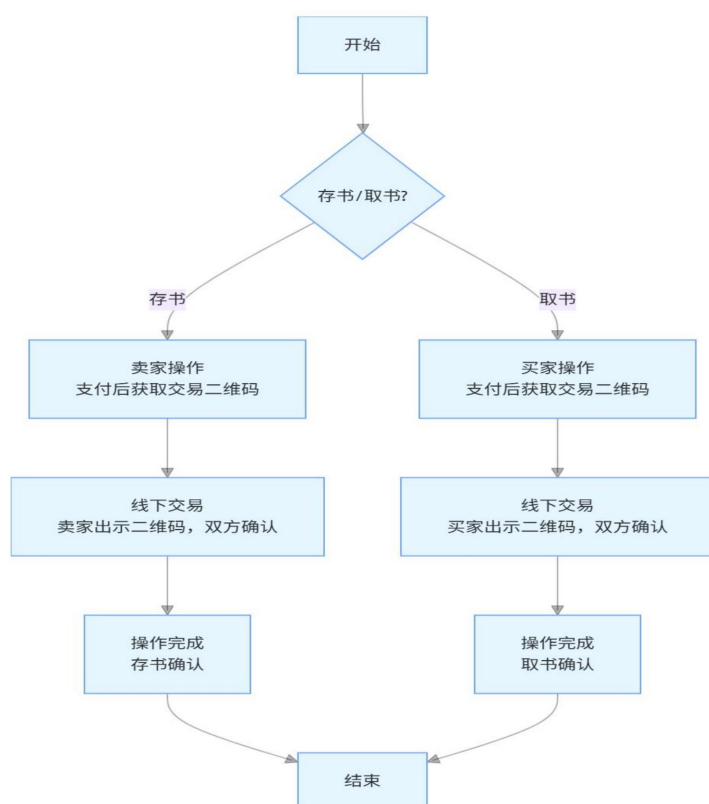
Show the QR code offline, and both parties confirm the completion of book deposit

(3) Book Retrieval Branch:

Buyers obtain the transaction QR code after payment

(4) Show the QR code offline, and both parties confirm the completion of book collection.

(5) End: Conclude the book deposit/retrieval process



Flowchart of Offline QR Code Transaction Code in Figure 63

7. Administrator Backend Management Process

(1) Login Verification: Administrators log in to the backend system using their account credentials to ensure operational permissions.

(2) Functional Shunting: Select different modules according to management requirements:

User Management: View the user list, perform blocking operations on abnormal accounts (such as those involved in order brushing or violations), and update the account status to "disabled".

Transaction Monitoring: Verify order disputes (such as damaged books, non-receipt of goods), and call the payment interface to execute a refund for orders confirmed to require a refund.

(3) Operational Closed Loop: After the key operations (blocking, refunding, repair reporting) of each functional module are completed, the process automatically ends, and the system records operation logs for easy traceability.

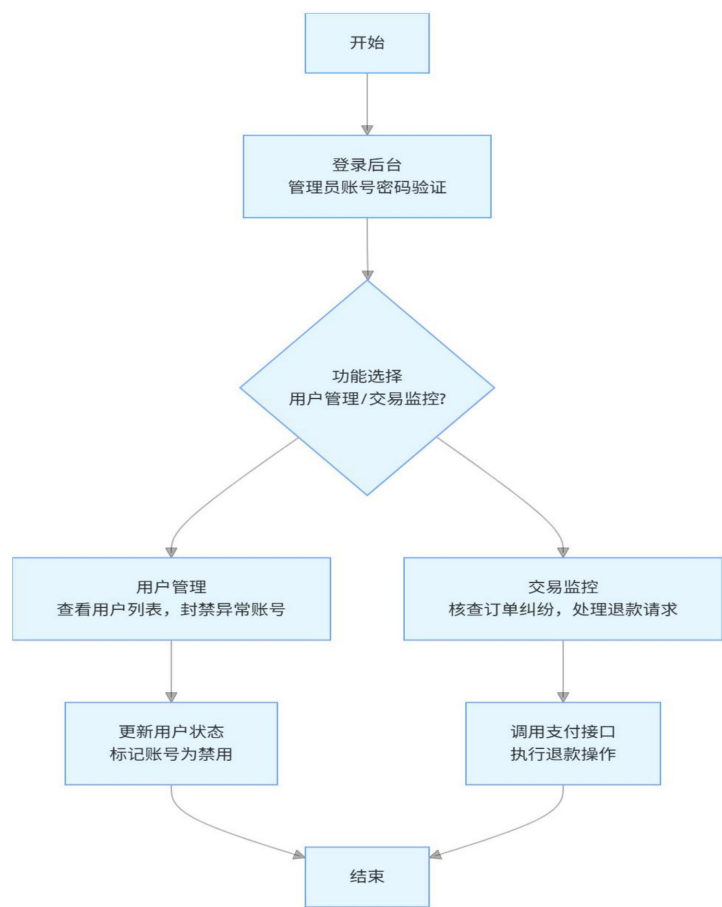


Figure 64 Flowchart of Administrator Backend Management Code

6.3.2 Key Part Code

1. Book Release Section

(1) ISBN Scanning and Auto-Fill

Simulate scanning the ISBN code and automatically populate the basic book information; use `setTimeout` to simulate the asynchronous scanning process and update the `isScanning` state to control the loading animation; update partial form fields through shallow copy (`...bookInfo`) to avoid resetting other input content.

```
const handleISBNScan = () => {
  setIsScanning(true);
  // 模拟ISBN扫描异步操作
  setTimeout(() => {
    setIsScanning(false);
    setBookInfo({
      ...bookInfo,
      isbn: "9787040396638",
      title: "高等数学（第七版）上册",
      author: "同济大学数学系",
      publisher: "高等教育出版社",
      edition: "第7版",
    });
  }, 2000);
};
```

Figure 65 ISBN Scan and Auto-Fill Code

(2) Image Upload and Management

Convert files to URLs and add them to the uploadedImages array, supporting multi-file selection; remove the image at the specified index through array filtering; use URL.createObjectURL to generate a temporary URL, which may need to be uploaded to the server in an actual project.

```
const handleImageUpload = (event: React.ChangeEvent<HTMLInputElement>) => {
  const files = event.target.files;
  if (files) {
    const newImages = Array.from(files).map((file) => URL.createObjectURL(file));
    setUploadedImages([...uploadedImages, ...newImages]);
  }
};

const removeImage = (index: number) => {
  setUploadedImages(uploadedImages.filter((_, i) => i !== index));
};
```

Figure 66 Image Upload and Management Code

(3) AI Audit Simulation

Control the review status (pending review / under review / approved / rejected) through the aiReviewStatus enum value; use a timer to simulate the processing time of AI review, and in actual projects, the backend API should be called.

```
const handleAIRewiew = () => {
  setAiReviewStatus("reviewing");
  // 模拟AI审核过程
  setTimeout(() => {
    setAiReviewStatus("approved");
  }, 3000);
};
```

Figure 67 AI Review Simulation Code

(4) Label Management

Check if the label already exists when adding it to avoid duplicates; use the filter method to implement label deletion and maintain the immutable data pattern.

```
const addTag = (tag: string) => {
  if (tag && !bookInfo.tags.includes(tag)) {
    setBookInfo({
      ...bookInfo,
      tags: [...bookInfo.tags, tag],
    });
  }
};

const removeTag = (tagToRemove: string) => {
  setBookInfo({
    ...bookInfo,
    tags: bookInfo.tags.filter((tag) => tag !== tagToRemove),
  });
};
```

Figure 68 Label Management Code

(5) Release Process Control

Use async/await to handle the publishing process, simulate API requests via Promise; convert string prices to numbers, calculate derived fields such as original price and carbon credits; simultaneously update the publishedBook and step states, triggering page switching.

```
const handlePublish = async () => {
  setIsPublishing(true);

  try {
    // 模拟发布过程
    await new Promise((resolve) => setTimeout(resolve, 2000));

    // 生成新的书籍记录（包含计算字段）
    const newBook = {
      id: `book_${Date.now()}`,
      ...bookInfo,
      price: Number(bookInfo.price),
      originalPrice: Math.round(Number(bookInfo.price) * 1.8),
      images: uploadedImages.length > 0 ? uploadedImages : [/* 默认图片 */],
      sellerId: "user_001",
      status: "available",
      views: 0,
      likes: 0,
      carbonPoints: Math.floor(Number(bookInfo.price) * 0.6),
      publishDate: new Date().toISOString().split("T")[0],
      location: "东区宿舍",
      publishTime: new Date().toLocaleString(),
      estimatedViews: Math.floor(Math.random() * 50) + 20,
      marketPrice: Math.round(Number(bookInfo.price) * 1.8),
    };

    setPublishedBook(newBook);
    setStep(5); // 跳转到发布成功页面
  } catch (error) {
    console.error("发布失败:", error);
  } finally {
    setIsPublishing(false);
  }
};
```

Figure 69 Release Process Control Code

2. QR Code Transaction Section

(1) QR Code Scanning and Verification

Simulate the process of a camera scanning, display a loading animation, and automatically populate test data; find matching transaction records based on the QR code content, and generate a verification result containing transaction details; in a production environment, this should call the backend API to perform real QR code verification.

```
const handleScanQR = () => {
  setIsScanning(true)
  // 模拟扫码过程
  setTimeout(() => {
    setIsScanning(false)
    setQrCodeInput("QR_TXN002_RECEIVE")
    handleVerifyTransaction("QR_TXN002_RECEIVE")
  }, 2000)
}

const handleVerifyTransaction = async (qrCode: string) => {
  try {
    // 实际应用中应该调用API验证二维码
    const transaction = myTransactions.find((t) => t.qrCode === qrCode)
    if (transaction) {
      setSelectedTransaction(transaction.id)
      setVerificationResult({
        transaction,
        book: { title: transaction.bookTitle },
        seller: { name: transaction.type === "购买" ? transaction.seller : "您" },
        buyer: { name: transaction.type === "出售" ? transaction.buyer : "您" },
      })
    }
  } catch (error) {
    console.error("验证二维码失败:", error)
  }
}
```

Figure 70 QR Code Scanning and Verification Code

(2) Transaction Completion Processing

Update the transaction to "completed" based on the transaction type and current status; display success prompt messages for different roles; record the specific time when the transaction was completed; in a real scenario, this should call the backend API to update the transaction status in the database.

```
const handleCompleteTransaction = (transactionId: string) => {
  // 找到对应的交易
  const transaction = myTransactions.find((t) => t.id === transactionId)
  if (!transaction) return

  // 根据交易类型显示不同的成功消息
  if (transaction.type === "出售" && transaction.status === "待交付") {
    setSuccessMessage("交付成功! 您已确认将书籍交付给买家, 交易即将完成。")
  } else if (transaction.type === "购买" && transaction.status === "待收货") {
    setSuccessMessage("收货成功! 您已确认收到书籍, 交易已完成。感谢您的环保行动!")
  }

  // 模拟更新交易状态
  const updatedTransactions = myTransactions.map((t) => {
    t.id === transactionId ? { ...t, status: "已完成", completedTime: new Date().toLocaleString() } : t,
  })

  // 3秒后清除成功消息
  setTimeout(() => {
    setSuccessMessage(null)
  }, 5000)
}
```

Figure 71 Transaction Completion Processing Code

(3) State Style Mapping

Map transaction statuses to corresponding color class names for visual differentiation of different statuses; ensure consistent visual representation of statuses throughout the application; easily add new statuses or modify the colors of existing statuses.

```
const getStatusColor = (status: string) => {
  switch (status) {
    case "待交付":
      return "text-orange-600"
    case "待收货":
      return "text-blue-600"
    case "已完成":
      return "text-green-600"
    case "已取消":
      return "text-red-600"
    default:
      return "text-gray-600"
  }
}

const getStatusBadgeColor = (status: string) => {
  switch (status) {
    case "待交付":
      return "bg-orange-100 text-orange-800"
    case "待收货":
      return "bg-blue-100 text-blue-800"
    case "已完成":
      return "bg-green-100 text-green-800"
    case "已取消":
      return "bg-red-100 text-red-800"
    default:
      return "bg-gray-100 text-gray-800"
  }
}
```

Figure 72 State Style Mapping Code

3. Carbon Credit Section

(1) Points Management System

Displays the user's total points, points this month, ranking, and carbon reduction through a card-style layout; implements a points-based level promotion mechanism, with a Progress Bar to intuitively show the upgrade progress; uses gradient background and icons to enhance visual effects, aligning with the environmental protection theme.

```
// 积分商城区域
<div className="grid grid-cols-1 md:grid-cols-4 gap-6 mb-8">
  <Card className="bg-gradient-to-r from-green-500 to-green-600 text-white">
    <CardContent className="p-6">
      <div className="flex items-center justify-between">
        <div>
          <p className="text-green-100">总积分</p>
          <p className="text-3xl font-bold">{userStats.totalPoints}</p>
        </div>
        <Leaf className="h-12 w-12 text-green-200" />
      </div>
    </CardContent>
  </Card>
  // ...其他统计卡片
</div>

// 等级进度条
<Progress value={({userStats.totalPoints % 500} / 5) className="h-3" />
```

Figure 73 Code of the Integral Management System

(2) Points Mall and Redemption System

Supports filtering products by prize category; select a prize and confirm redemption, choose a delivery method (self-pickup or dormitory delivery), confirm points consumption and remaining points, generate a unique redemption code and expiration date, display a success prompt and usage instructions; manage the redemption process through dialog components to ensure a smooth user experience.

```
const handleRedeem = (reward: any) => {
  if (userStats.totalPoints >= reward.points) {
    setSelectedReward(reward)
    setShowRedeemDialog(true)
  }
}

const handleConfirmRedeem = () => {
  setRedeemLoading(true)

  // 模拟兑换过程
  setTimeout(() => {
    // 生成兑换记录
    const record = {
      id: `exchange_${Date.now()}`,
      rewardName: selectedReward.name,
      points: selectedReward.points,
      deliveryMethod: deliveryMethod,
      exchangeCode: `EX${Math.random().toString(36).substring(2, 10).toUpperCase}`,
      validUntil: new Date(Date.now() + 30 * 24 * 60 * 60 * 1000).toLocaleDateString(),
      remainingPoints: userStats.totalPoints - selectedReward.points,
      // ...其他信息
    }

    setRedeemRecord(record)
    setRedeemLoading(false)
    setShowRedeemDialog(false)
    setShowSuccessDialog(true)
  }, 1500)
}
```

Figure 74 Code for the Points Mall and Redemption System

(3) Points History and Leaderboard

Display the user's points acquisition records in a timeline format, including transaction type, book information, and points earned; show users with leading points, with the top three highlighted using special styles, and the user's own ranking displayed separately; distinguish points acquisition types and ranking positions through different colors and icons.

```

// 积分历史记录
{pointsHistory.map((record) => (
  <div key={record.id} className="flex items-center justify-between p-4 border rounded-lg">
    <div className="flex items-center gap-4">
      <div className="w-10 h-10 bg-green-100 rounded-full flex items-center justify-center">
        <Leaf className="h-5 w-5 text-green-600" />
      </div>
      <div>
        <h4 className="font-medium">{record.type}</h4>
        {record.bookTitle && <p className="text-sm text-gray-600">{record.bookTitle}</p>
        <p className="text-xs text-gray-500">{record.date}</p>
      </div>
    </div>
    <div className="text-right">
      <p className="font-bold text-green-600">+{record.points} 积分</p>
      {record.carbonSaved > 0 && <p className="text-xs text-gray-600">减碳 {record.carbonSaved}kg</p>}
    </div>
  </div>
)}

// 排行榜展示
{leaderboard.map((user) => (
  <div key={user.rank} className="flex items-center justify-between p-4 border rounded-lg">
    <div className="flex items-center gap-4">
      <div
        className={`w-10 h-10 rounded-full flex items-center justify-center font-bold text-white ${
          user.rank === 1 ? "bg-yellow-500" : user.rank === 2 ? "bg-gray-400" : "bg-orange-500"
        }`}
      >
        {user.rank <= 3 ? <Trophy className="h-5 w-5" /> : user.rank}
      </div>
      {user.rank <= 3 ? <Trophy className="h-5 w-5" /> : user.rank}
    </div>
    <div>
      <h4 className="font-medium">{user.name}</h4>
      <Badge variant="outline" className="text-xs">{user.level}</Badge>
    </div>
    <div>
      <div className="text-right">
        <p className="font-bold text-green-600">{user.points} 积分</p>
      </div>
      <div className="flex items-center gap-1">
        {[...Array(5)].map((_, i) => (
          <Star key={i} className={`h-3 w-3 ${i < Math.floor(user.points / 500) ? "fill-yellow-400" : "text-gray-300"}`
        ))}
      </div>
    </div>
  </div>
)}

```

Figure 75: Code for Integration History and Leaderboard

6.4 User System Prototype Interaction and Feedback

6.4.1 User Interview

1. Interview Content

The interviewees for our system interview were 1 undergraduate student currently studying in Tianjin and 1 undergraduate student from outside Tianjin. Before the formal start of the interview, we informed the users about the development stage of the system and explained the user-expected functions that the system aims to achieve.

After the user has a preliminary understanding and enters the formal interview, we log into the system and then demonstrate all the functions of the system to the user as a new user, including registration, login, book screening, book search, related book recommendation, book posting, QR code transaction, carbon credits, and personal information management. During the demonstration, we explain to the user one by one the information and identifiers presented on the system pages, including the website logo, the structure of each page, the division of information list areas, the meaning of information, etc. After the demonstration, we re-enter the system's registration page, and the user conducts a preliminary trial. During this process, the developer assists the user and answers any questions that arise during the user's operation. After the user has tried out the system, the developer conducts an interview according to the outline and, with the user's consent, records the entire interview process.

2. Interview Outline

- (1) To what extent do you think the current system's functions have resolved the initial difficulties you encountered in buying or selling books?
- (2) We have made multiple improvements based on the feedback collected during the prototype communication phase, and also made revisions and refinements due to the unsuccessful delivery last time. Have you noticed these improvements, and how do you feel about them now?
- (3) From the perspectives of interface aesthetics and functional usability, what are your evaluations of the current system's user experience?
- (4) During the use of the system, have you ever encountered situations such as laggy pages, delayed operations, or untimely function responses?
- (5) For the currently delivered system, what areas do you think still need improvement? Do you have any suggestions for new features?

6.4.2 User Feedback

After interviewing users, they provided us with timely feedback, giving suggestions for improvement:

Improvement: In some secondary interfaces, you need to click the back button twice to return to the first interface.

Problem	Solution
The information prompt in the chat window is not obvious	Continue iterative optimization to make the prompt box more prominent
Interface algorithm and front-end optimization, interaction optimization	Continue iterative optimization
Conduct customized recommendation based on user profiles	will design a similar page that appears on the home page when a new Weibo user logs in Selection of personalized and interesting content, and subsequent actions will be based on the selection Line Recommendation
Add the function of wish list	Add a wish list feature module
Offline book collection navigation function	The buyer and seller shall negotiate the transaction location by themselves and use QR codes for the transaction

	Offline transaction
Users can provide feedback or suggestions regarding their experience with this transaction	Add evaluation module
Evaluation System	

Table 8 User Interaction Problem Feedback

6.5 System Interface Display

1. Home Page

This is the homepage of the "Campus Book Recycling" platform, positioned as a campus second-hand book trading platform that advocates "circulating knowledge and making the Earth greener", combining environmental protection and cost-saving values. The page features a search bar that allows users to search by book title, author, and ISBN, as well as a "Popular Categories" section. The top navigation bar includes functional entrances such as the homepage, post books, QR code trading, carbon credits, login, and personal center, facilitating users' second-hand book trading and related operations.



Figure 76 System Homepage

This is the featured books display page of the "Campus Book Recycling" platform, focusing on second-hand book transactions. At the top of the page, there is a "Featured Books" title and filtering buttons, and below are category labels such as "All, Textbooks, Reference Books, Novels". Multiple second-hand books are displayed, each with information including a cover image, condition indicator (e.g., 90% new), book title (including version), author/publisher, rating, carbon credits, price (original price and selling price), seller, and book category (e.g., East Dormitory, Textbook, etc.), making it convenient for users to select suitable second-hand books.



Figure 77 System Homepage

This is the usage guide and related information page for the "Campus Book Recycling" platform, with the core being the "How to Use" section. Through three steps of "Publish a Book", "Find a Book", and "Earn Points", paired with icons, it clearly guides users through the operation process. Below is the platform footer, which includes the "Campus Book Recycling" logo, philosophy, as well as information on "Quick Links" (Publish a Book, Search for Books, Carbon Points), "Help Center", and contact details.



Figure 78 Bottom of the System Homepage

This is the process page for publishing second-hand books on the "Campus Book Recycling" platform, which is used to assist users in publishing idle second-hand books. The title "Publish Second-Hand Books" clearly indicates the function, and the "Publishing Progress" consists of a total of 5 steps, including the "Basic Information" section, as well as "Upload Images", "AI Review", "Publish Confirmation", and "Publish Success". Step 1 requires "Filling in Basic Information", supports scanning the ISBN code, and clicking the "Scan ISBN" button to scan the code on the back of the book to automatically fill in the information.

You can also manually fill in information such as ISBN, book category, book title, author, etc., to guide users through the initial information entry for second-hand book posting.



Figure 79 Second-hand Book Publishing Interface

This is the QR code transaction page of the "Campus Book Recycling" platform, with tabs for "My Transactions", "Scan to Verify", and "User Guide" at the top. Under "My Transactions", transaction records are displayed. Taking "Advanced Mathematics (7th Edition)" as an example, information such as the buyer, transaction location, creation/expiration time, price, etc. is presented, and it also includes the transaction QR code as well as copy and download functions, facilitating users to complete the verification and delivery processes of book transactions.

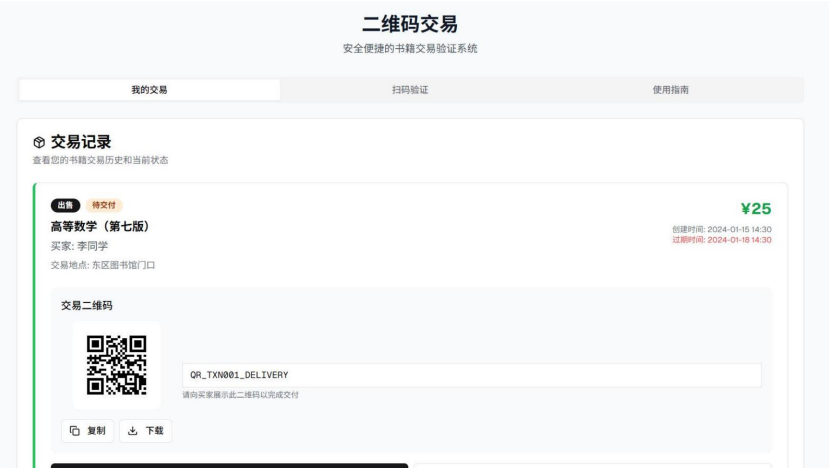


Figure 80 QR Code Transaction Interface

This is the QR code transaction - scan verification page of the "Campus Book Recycling" platform. The page is divided into tabs of "My Transactions", "Scan Verification", and "User Guide", and currently it is on "Scan Verification". The left side is the "Scan Transaction QR Code" area, where you can click "Start Scanning" to scan with the camera or manually enter the verification code for verification; the right side is the "Verification Result" area, which displays the result after scanning/entering the code, used for verifying the transaction QR code in second-hand book transactions to ensure transaction security and standardization.



Figure 81 Scan Code Verification Interface

This is the QR code transaction usage guide page of the "Campus Book Recycling" platform. The page has tabs for "My Transactions", "Scan for Verification", and "Usage Guide", and currently it is on the "Usage Guide". The content is divided into "Seller Operation Process" (Publish

Seller's operation process (book → generate QR code → schedule transaction → complete delivery), "Buyer's operation process" (browse and place order → pay order → schedule transaction → scan QR code to receive goods), and also lists "Safety reminders" (conduct transactions on time, verify the amount, confirm accuracy before proceeding with the transaction, and contact customer service if any issues arise), guiding users to complete second-hand book transactions in a standardized and secure manner using QR codes.



Figure 82 Guide Interface

Carbon Credit Center Page: Displays the core data of users' carbon credits, including total credits, credits this month, ranking, and carbon reduction amount. Presents the current level, sets up entrances to the credit mall, credit records, and leaderboard, encouraging users to exchange environmental protection actions for rewards.



Figure 83 Carbon Credit Center Interface

Points Mall Page: It belongs to the Carbon Points Center section, where users can redeem prizes with their points. There are categories such as dining (e.g., Starbucks coffee vouchers), shopping (campus bookstore vouchers), supplies (eco-friendly tote bags), services (library extension vouchers), etc., clearly displaying the points required for each prize and its inventory, facilitating redemption.

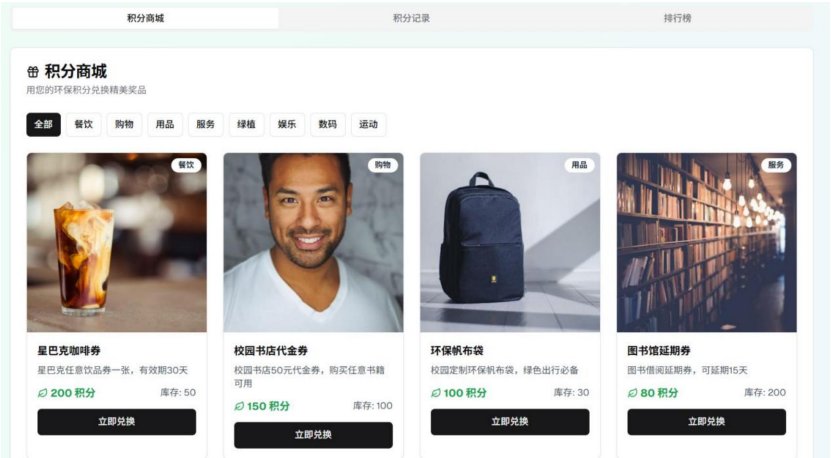


Figure 84: Points Mall Interface

Points Record Page: Records the user's history of points acquisition and usage, such as points and carbon reduction generated from completed transactions (e.g., transactions for Advanced Mathematics, Java Programming Concepts), first-time releases (e.g., Linear Algebra), etc., to facilitate users in tracing the dynamics of their points.



Figure 85 Points Record Interface

Ranking Page: Displays the ranking of campus environmental protection experts, listing user names, levels, and points, such as Student Zhang (Green Pioneer, 2850 points), Student Li, etc., to motivate users to participate in environmental protection transactions for ranking boost.

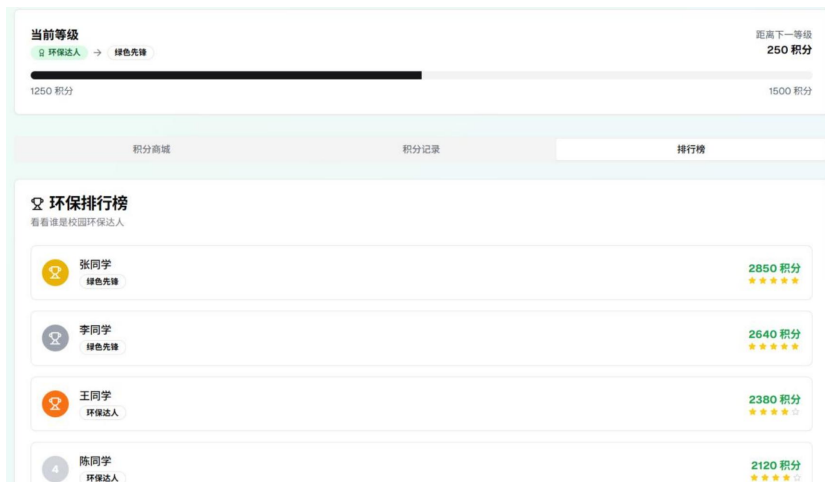


Figure 86: Interface of the Points Ranking List

Personal Center - Account Settings Page: Manage account information and transaction records, set up notifications for transactions, points changes, and system messages, adjust privacy (display real name, contact information) and security (change password, bind WeChat) settings, and display user basic information (name, level, contact information, etc.) and Statistical Data (total points, number of transactions, credit score) on the left side



Figure 87 Personal Center Interface

7 System Testing

7.1 System Testing Method

1. Functional Testing

(1) Integration Testing: Verify the interfaces and data interactions between system modules to ensure error-free data transmission for functions such as search, display, and transaction; check database operations to ensure data accuracy, redundancy-free, and security.

(2) System Testing: Conduct end-to-end testing of functions such as user registration, book selection, transaction, and evaluation to ensure that all components work together to meet business requirements.

2. Acceptance Testing: Simulate real usage scenarios to verify whether the system functions meet user requirements and business specifications.

3. User Experience Testing

(1) Usability Testing: Evaluate whether the website navigation, layout, and Interaction Design are user-friendly, and whether the operation process is convenient.

(2) User Testing: Invite real users to experience, collect feedback on aspects such as interface, operation, and functionality, and optimize User Experience.

7.2 System Testing Environment and Tools

1. Local Development Environment: Developers use personal computers to write and debug code, conduct unit tests, such as checking whether the data call on the book details page is correct, and quickly fix basic code issues.

2. Test Environment: The pre-release environment simulates the online environment, testing the integration of functional modules, such as the entire transaction process
3. Unit Test Framework: Detects book classification algorithms, database connections, front-end interaction logic, etc., to quickly identify code logic errors.

7.3 Black Box and SAST White-Box Testing

7.3.1 Login

1. Black box testing

Input operation	Output Result
Enter a valid 11-digit mobile phone number → Switch to password login → Enter a 6-digit password → Click "Login"	The button displays "Logging in...", and after 2 seconds, it redirects to the home page
Enter an 11-digit mobile phone number → Switch to verification code login → Click Click "Send Verification Code"	Console outputs "Send verification code to [mobile number]"
Click the eye icon on the right side of the password input box	The password input box toggles between plaintext and ciphertext, with the icon synchronizing accordingly. Change
Click the "WeChat Login" tab → Click "WeChat Login Record" button	Console outputs "WeChat Login"
Click "Login" without entering any content	Login button is disabled, no redirection
Enter a 10-digit mobile phone number → Try to click "Send Verification Verification Code "	Button disabled (format validation needs to be added)
Enter a mobile phone number containing letters → Switch to password login → Enter Enter a 5-digit password → Click "Login"	Button disabled (password strength verification needs to be added)

Click the "Password Login" tab → Switch to "Verification Code Login Tab	The verification code input box area is displayed, and the password input box is hidden
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Table 9 Black Box Testing of Login Function

2. SAST white-box testing

- (1) State Transition: When showPassword is toggled, whether the type attribute of the input box is synchronously changed (password/text); whether the isLoading state correctly controls the button's disabled state and text switching.
- (2) Function Logic: Whether the asynchronous redirection logic in handlePhoneLogin is executed (setTimeout and window.location); whether sendVerificationCode correctly retrieves the phoneNumber parameter and outputs it.
- (3) Component Rendering: Whether TabsContent conditionally renders the corresponding form based on the value property (e.g., the sms tab displays the verification code input box); whether the form values and states (phoneNumber/password) are two-way bound.
- (4) Password Show/Hide: When clicking the eye icon, the type attribute of the input box switches correctly (password, text); the icon style updates synchronously (Eye, EyeOff).
- (5) Login process: After clicking the "Login" button, the isLoading state is set to true, and the button displays "Logging in..."; after 2 seconds, the simulated request is completed, and the page redirects to the home page.
- (6) Verification code function: After entering the mobile phone number, the "Send verification code" button can be clicked; after clicking, the sendVerificationCode function is called, and the mobile phone number is output to the Console (simulation only).
- (7) Tab switching: When switching between "Phone Login / WeChat Login" or "Password Login / Verification Code Login", the corresponding content area is correctly displayed / hidden.

7.3.2 Registration

1. Black box testing

Input operation	Output Result
Enter "13800138000" (correct) and "1234567890" (Error)	Correct format allows input, incorrect format prompts
Enter "123456" (correct) and "123" (incorrect)	Passwords with 6 or more characters are allowed; those with fewer than 6 characters will prompt a

	message
Enter the correct mobile phone number → Click "Send Verification Code"	The button displays a countdown, and the Console outputs the mobile phone number
Click the "Next"/"Previous" button to navigate through the 3 steps Switch	Correctly jump to the corresponding step, and the Progress Bar updates synchronously
Select the "Computer Science" major in Step 2 and " Freshman Year	Dropdown options are correctly displayed and selected
Click "Complete Registration" directly without checking the "Terms of Service"	Prompt "Please agree to the Terms of Service"
After completing all steps and successfully registering	Show points rewards, badges, and achievement prompts
Enter password "123456", confirm password "123457"	Prompt "The two passwords do not match"

Table 10 Black Box Testing of Registration Function

2. SAST white-box testing

(1) Multi-step state transition: When step switches from 1 to 3, whether the form elements (such as student ID, major) of the corresponding step are correctly rendered; when step validation fails, whether the error message is accurately located to the corresponding field (such as incorrect mobile phone number format).

(2) Two-way binding of forms: When entering a name/email, whether the corresponding fields in formData are updated in real time; when checking the agreement checkbox, whether the states of agreeTerms/agreePrivacy are synchronously changed.

(3) Countdown mechanism: After clicking "Send Verification Code", does the countdown start decrementing from 60; after the countdown ends, does isCodeSent automatically reset to false?

(4) Mobile phone number verification: When entering "13800138000" (correct), does validateStep1 return without error; when entering "1234567890" (incorrect), is the prompt "Please enter a valid mobile phone number" triggered?

(5) Password consistency check: When the password matches the confirmed password, does validateStep1 pass; when entering "123456" and "123457", is the prompt "The two passwords do not match" displayed?

- (6) Required Agreement Validation: When "Terms of Service" is not checked, does validateStep3 prevent proceeding to the next step? When both agreements are checked, is it allowed to enter the registration process?
- (7) Registration Request Simulation: In handleRegister, is setTimeout correctly simulating a 2-second delay? After successful registration, does registrationData contain all required fields (such as id, name, points)?
- (8) Error handling: When simulating registration failure, does errors.general correctly display the "Registration failed" prompt; when there is a network exception, does it prevent user data from being written to localStorage?

7.3.3 Publish Books

1. Black box testing

Input operation	Output Result
Click the "Scan ISBN" button to simulate a successful scan	Automatically populate information such as ISBN, title, author, etc.
Click directly without filling in required fields such as book title / price, etc. " Next Step	Form prompt for required field error
Select multiple images to upload, and click to delete the uploaded images Button	The image was successfully uploaded and displayed, and the deletion function is working properly
Click the "Start AI Review" button and wait for the simulated review Completed	Displays the status of being under review, and finally prompts that the review has passed
Check the book information preview on the release confirmation page	Preview information is consistent with the filled content
Click the "Confirm Release" button and wait for the simulated release to complete	Jump to the publication success page and display the book ID
Click buttons such as "View Book Details" and "Continue Publishing"	The button triggers the corresponding function, such as jumping to the details page
Press Enter to add input labels, click the delete button on the label to	The tag was successfully added to the list and can be deleted

Button	
Input non-numeric characters or negative numbers as the price	Form prompts that the price format is incorrect

Table 11 Black Box Testing of Book Publishing Function

2. SAST white-box testing

(1) Step switching condition:

Step 1→2: When clicking "Next" without filling in the book title / price, does it prevent the switch and prompt for required fields?

Step 2→3: When no image is uploaded and the "Next" button is clicked, does the image upload prompt appear?

Step 3→4: When AI review fails, click "Next", whether to block the switch;

(2) Scan Status Management: After clicking "Scan ISBN", whether isScanning is set to true and the button is disabled; after scanning is completed, whether isScanning is reset to false and the button becomes clickable again

(3) Review Status Update: After clicking "Start AI Review", does aiReviewStatus change to "reviewing" → "approved" in sequence; during the review process, does the status remain unchanged when the page is refreshed (currently not handled, local storage needs to be added)

(4) ISBN Scan Simulation: Whether setTimeout in handleISBNScan correctly delays execution by 2 seconds; whether fields such as ISBN and book title in bookInfo are correctly populated after scanning

(5) AI Review Simulation: Whether aiReviewStatus is correctly updated to "approved" after a 3-second delay in handleAIReview; whether carbon credits (price * 0.6) are correctly calculated after the review is approved

(6) Publishing Logic: Does handlePublish generate a book containing fields such as id, carbonPoints, etc.?

Record; after successful publication, whether it correctly jumps to Step 5 and displays the publication record.

7.3.4 Carbon Credits

1. Black box testing

Input operation	Output Result

View data such as total points, points for this month, ranking, carbon reduction amount, etc. As shown	Data is consistent with the preset value, and the unit display is correct
View the level progress bar and the points prompt for the next level	The Progress Bar correctly reflects the current points ratio, and the prompt information is accurate Sure
Click the "All" and each category button (e.g., Dining, Supplies) etc.	Correctly filter the corresponding classified prizes, and the button state switches
Select a prize with a point requirement higher than the current points	The redemption button displays "Insufficient points" and cannot be clicked
Select redeemable prize → Confirm redemption → Select collection method →Confirm Exchange	Pop up the successful redemption dialog box, displaying the redemption code
Switch between self-pickup and delivery methods in the redemption dialog box	The corresponding information form for the delivery method is correctly displayed / hidden
After successful redemption, click "Copy redemption code" and "Share" Button	Redemption code successfully copied, sharing function triggered
Switch to the "Points Record" tab to view points acquisition History	Correctly display historical records, including books, points, and dates
Switch to the "Leaderboard" tab to view your personal ranking and Top users	Ranking information is correctly displayed, and the current user's position is highlighted

Table 12 Black Box Testing of Carbon Credit Function

2. SAST white-box testing

(1) Classification and filtering status: When clicking the "Food and Beverage" classification button, does selectedCategory update to "Food and Beverage", and does filteredRewards correctly filter the corresponding rewards?

(2) Redeem dialog state: When clicking the "Redeem Now" button, is showRedeemDialog set to true, and is selectedReward correctly passed in?

- (3) Delivery method status: When switching between self-pickup / delivery methods, whether deliveryMethod is correctly updated, and whether the delivery information form is linked to display / hide.
- (4) Points verification logic: When calling handleRedeem, whether the redemption button is correctly disabled and the "Insufficient points" prompt is displayed when points are insufficient.
- (5) Redemption process simulation: Does the setTimeout in handleConfirmRedeem correctly simulate a 1.5-second delay, and does the generated redeemRecord contain all required fields?
- (6) Redeem Code Function:
- Does copyExchangeCode correctly call navigator.clipboard.writeText, and does the sharing function trigger log output?

7.3.5 QR Code Transaction

1. Black box testing

Input operation	Output Result
Click on the "My Transactions"/"Scan Code Verification"/"User Guide" tab	Correctly switch content area, label highlights
View the transaction list in the "My Transactions" tab	Status label colors are correct (Pending Delivery / Completed)
View the QR code area of a single transaction	Correctly generate QR code patterns and text
Click the "Start Scan" button and wait for the simulated scan to complete	Automatically fill in the verification code and display the verification result
Enter "QR_TXN002_RECEIVE" in the input box and click "Verify"	Display the verification results of the corresponding transaction
Click the "Confirm Delivered" or "Confirm Received" button	Display a success message, and update the transaction status to completed
View the seller/buyer operation steps on the "User Guide" tab	The step instructions are clear, and the icons correspond to the text
Click the expandable item of "Frequently Asked Questions"	Correctly display the problem details, with smooth interaction

Table 13 Black Box Testing of QR Code Transaction Function

2. SAST white-box testing

(1) Tab state transition: The default value of selectedTab is "transactions", and it updates correctly when a tab is clicked (e.g., the UI refreshes synchronously when switching to "scan"); when the URL parameter transactionId changes, selectedTransaction automatically updates and locates the corresponding transaction.

(2) QR code scanning status control: When isScanning is set to true, the scanning animation is displayed; after 2 seconds, the preset QR code is automatically filled in and verification is triggered; after scanning is completed, isScanning is reset to false, and the button returns to a clickable state.

(3) Verification Result Status: verificationResult correctly stores transaction data after barcode scanning / manual verification; the success message successMessage automatically clears after being displayed for 5 seconds.

(4) Scan code simulation logic: setTimeout in handleScanQR is delayed by 2 seconds to execute and correctly populate

qrCodeInput is "QR_TXN002_RECEIVE"; clicking other buttons during the simulated QR code scanning process will not interrupt the state (debounce handling needs to be added).

(5) Transaction verification logic: handleVerifyTransaction matches records in myTransactions based on qrCode; when there is no matching record, the error is not handled (currently it returns directly, and prompt logic needs to be added).

(6) Transaction completion logic: handleCompleteTransaction updates the transaction status to "completed" and records the time; success messages display different text based on the transaction type (sale/purchase).

7.4 Environmental Testing

The compatibility test of the system runs well on browsers (mainly Google), with good response speeds under different networks, and the screen resolution and display effect meet the system's expectations.

(1) Test devices: laptop, mobile phone;

(2) Network testing scenario:

High-speed broadband connection, medium-speed Wi-Fi, intermittent network connection, offline mode (check offline cache function).

8 System Delivery

8.1 Delivery Process Records

User 1's delivered video:

User 2's delivered video:

8.2 User Feedback During the Delivery Process

1. Advantages reported by users:

- (1) The carbon credit function is very appealing, which can arouse some of the users' purchasing desire and also provide users with a different experience in the system beyond just buying books.
- (2) Basically solved the pain point of the poor transaction of second-hand books.
- (3) The home page is quite user-friendly, enabling users to quickly get the hang of the operating system.

2. Disadvantages reported by users:

- (1) The names and images of the books have not been fully matched, which affects the visual experience.
- (2) The interface design is relatively simple and lacks options such as personalized skins.

3. New pain points/requirements raised by users:

New pain points/requirements	Solution
Add Wishlist Function	Gradually designed and added in subsequent development

Table 14 User New Requirements Table

8.3 User Delivery Receipt

1. User 1

交付回执单

交付时间: 2015年6月13日

交付对象(目标用户):

交付步骤: 系统介绍, 演示, 指导用户实际操作

交付后进行访谈调查

用户评价(满意度): 9.7分, 体验感很好, 能满足需求, 功能也很齐全, 为校园生活提供便捷.

用户签字:

Figure 88 User 1 Delivery Receipt

2. User 2

交付回执单

交付时间: 2015年6月13日 16点, 2分

交付对象(目标用户):

交付步骤: 系统介绍, 演示, 指导用户实际操作

交付后进行访谈调查

用户评价(满意度): 9.3分, 功能很实用, 界面设计新颖.

用户签字:

Figure 89 User 2 Delivery Receipt

9 Contribution

The specific division of labor and implementation are as shown in the following table.

Name	Division of Labor	Implementation Status
Ni Zixuan	- Usually participates in each experiment seriously, responsible for part of the image drawing and description. - Oversee the project direction, formulate clear development processes and stage goals, and ensure the orderly progress of the project. Connect team members and coordinate design, development, and testing. - and other links. - Participate in core function testing, troubleshoot page vulnerabilities and interaction flaws, and ensure a smooth platform experience.	has been fully completed

	4. Participated in the writing of the final overall report.	
Ma Junyan	1. Usually participates in each experiment seriously, responsible for part of the image drawing instructions and user interview communication. 2. Participated in project development discussions, providing creative ideas and page layout suggestions for the QR code transaction module. 3. Participate in core function testing, troubleshoot page vulnerabilities and interaction flaws, and ensure a smooth platform experience. 4. Participated in the writing of the final overall report.	has been fully completed
Tao Yuming	1. Usually participate in each experiment seriously, responsible for the drawing of each task board and the integration and writing of the overall report. 2. Participated in project development discussions and contributed creative ideas to features such as carbon credit visualization and redemption mall gameplay. 3. Participate in core function testing, troubleshoot page vulnerabilities and interaction flaws, and ensure a smooth platform experience. 4. Participated in the writing of the final overall report.	has been fully completed

Table 15 Team Division Execution Status Table

10 Summary

10.1 Problems Currently Existing in the System

10.1.1 Insufficient Functionality

1. Existing functions of the system

The existing functions can basically meet the needs of users, but there are still imperfections. After redeeming prizes in the carbon credit function module, the credits cannot change in a timely manner, and corresponding transaction records, transaction bills, etc., cannot be generated. In the future, the design of the database needs to be enhanced to provide backend support for the storage of carbon credits; in the registration module, after registering a new user, it is not possible to quickly log in to the homepage but instead stay on the registration success page, and it is also impossible to accurately log in during the login process.

After the account is registered, some operations are based on simulated accounts entering the homepage. During subsequent development and optimization, the login and registration module will be adjusted, a user data management library will be established, and complete user login and registration functions will be implemented. Regarding the function of immediately purchasing books, it cannot generate the corresponding payment QR code, nor can it quickly generate the corresponding transaction records. The subsequent system optimization will ensure that transaction records appear in the user's personal center interface in a timely manner, and the transaction QR code will subsequently establish a bridge with payment methods such as WeChat and Alipay.

2. The system currently has no functionality

The wishlist feature is a new feature that users hope to add, and it has not been specifically implemented yet. Functions such as algorithmic recommendation for the wishlist and algorithmic recommendation for users after entering the homepage have also not been implemented. In subsequent development and optimization, features will be gradually added to make the system more excellent.

10.1.2 Design Deficiency

The design interface of this part is not yet fully developed and requires further refinement.

1. The user personal center interface can be designed to be more aesthetically pleasing and eye-catching, making users more eager to experience our system and increasing their willingness to use it.

2. Optimize the images in the book to ensure a more perfect match between the book and the images, providing users with a better experience.

3. The interface design of the homepage does not incorporate elements of campus youth vitality, and the interface style is relatively monotonous. Subsequently, corresponding personalized design features will be provided to allow users to choose the style of logging into the system themselves.

10.1.3 Insufficient Data

In the Big data environment, data is characterized by large volume, high velocity, diversity, and low value density. System data fails to be updated in a timely manner and comprehensively covered, preventing book information from being promptly reflected in the system.

Real-time performance: The update of book source status lags behind. After a book is sold, the platform fails to synchronize information in a timely manner and still displays it in the recommendation list or search results, leading to users frequently clicking on invalid links; when sellers modify key information such as price and transaction method, the page data is refreshed with a delay, which is likely to trigger transaction disputes. In addition, there is a delay in batch

uploading of user behavior data (such as wishlist additions and browsing history), which affects the real-time recommendation effect.

Comprehensiveness: The dimensions of book information collection are single, only including basic attributes such as book title, author, and price, lacking key attributes such as ISBN number, whether it includes a CD / notes, and key chapter markings; the association between books and courses (such as corresponding major, grade, and course name) is not recorded, making it difficult to meet students' demands for precise screening based on academic needs. Meanwhile, user data lacks extended information such as interest tags and reading preferences, making it impossible to fully depict user profiles.

Aggregation: The data is only limited to the platform of this school, without integrating resources from surrounding universities or off-campus second-hand book markets, resulting in a limited total number of book sources; it is not linked with the borrowing data of the school library or the course textbook list of the educational administration system, making it impossible to obtain external information such as popular textbooks and recommended reading lists. In addition, the book purchase request information from campus communities, forums, and other channels has not been connected, leading to low efficiency in supply-demand matching.

Intelligence: The recommendation algorithm is simple and crude, failing to analyze users' search keywords and review content by combining deep learning or natural language processing technologies, thus unable to understand the potential needs behind semantics; lacking an intelligent early warning mechanism, it cannot predict the supply and demand fluctuations during the textbook exam season based on historical transaction data to assist sellers in stocking up in advance or users in purchasing books in a timely manner. Meanwhile, it does not utilize AI to automatically review book source information, making it difficult to quickly identify issues such as false descriptions and duplicate postings.

10.2 Future Outlook

After gaining a deeper understanding of user needs, we expect to add a wishlist function (which can quickly match the desired books) and a customized recommendation function.

1. **Wishlist Function:** Initially, it is envisioned to provide a prominent "Wishlist" entry on the User Interface, which can be set as an icon or menu option on the home page or in the personal center. The designed wishlist page will display information about the user's added wishlist books, including the book title, author, cover image (if available), etc., and can adopt a list layout. It also provides operation buttons for each wishlist book, such as "Edit", "Delete", and "Contact Seller", etc.

2. **Customized Recommendation:** When entering the homepage, book recommendations are centered around user profiles, which are constructed by collecting behavioral data such as

user browsing, purchasing, and searching. On this basis, a hybrid recommendation algorithm is applied. For new users, content-based recommendation is used to recommend popular books in their professional or interest fields; for old users, collaborative filtering is combined to display books favored by similar users from the same school and major. Additionally, the association rule algorithm can be utilized to recommend related books based on users' historical purchase records. For example, users who have bought a certain textbook will be recommended the corresponding supplementary study materials, thereby achieving accurate customized recommendations. For books on the wish list, a content-based recommendation algorithm is adopted to analyze features such as the title, author, and subject of the books on the wish list. When newly listed books have a high degree of match, they will be automatically recommended to users; in combination with the collaborative filtering algorithm, by referring to the wish lists and transaction records of other users with similar interests, similar books will be additionally recommended.

10.3 Course Achievements

: After learning this course, I have gained a great deal. Previously, I thought system development was very mysterious, but only after completing the course did I realize that from the initial communication with users to understand requirements, to designing functional modules, writing code for implementation, and then to repeated testing and optimization, every step

All are important. In the course, working on projects in a group has taught me teamwork. Group members have clear division of labor, communicate seriously about progress and solve problems every day, and their communication and cooperation abilities have improved significantly.

Hands-on practice has enabled me to apply the knowledge from books. Previously, I only learned theory from textbooks, but in actual development, issues such as code errors and function incompatibilities kept emerging. I had to search for information and debug on my own to solve these problems, gradually gaining a deeper understanding of the processes and technologies involved in system development. Now, I can learn to use AI to assist me in independently completing some functions of small systems, and I even dare to try incorporating new ideas into development, which has also exercised my innovative thinking.

: In this course, I was deeply involved in the entire process of system development. From accurately identifying pain points in campus second-hand book trading and environmental protection needs through demand research, to using AI-assisted functions in design, such as leveraging AI to streamline the logic of modules like book posting, trading, and carbon credits, I efficiently advanced the development. During the process, I coordinated team division of labor, coordinated development, testing, and other aspects, resolved issues such as AI tool adaptation

and functional conflicts, and honed my project management and communication skills. At the same time, I also learned a great deal, mastering techniques for using AI to optimize code and simplify processes, thereby improving development efficiency. Through system launch and iteration, I understood the key to translating information systems from theory to implementation and also recognized the potential of AI in empowering development.

Teenagers are always very idealistic, like newborn calves not fearing tigers. From envisioning perfect system functionality to making compromises during the development process, I have also come to understand the difficulty and hardship of the development process, but I am even more determined to bravely scale new heights. After the course ends, I will continue to optimize my own system and use technology to promote the recycling of campus resources.

: Taking this course has left a deep impression on me and brought me a great deal of gains. Initially, I found this course unfamiliar and troublesome, and I even had to rush to complete assignments during the lab sessions. However, after listening to the teacher's analysis of the course, I found it very reasonable and gradually accepted it, then wholeheartedly devoted myself to the subsequent system development. Although I encountered many setbacks, with the leadership of the team leader and the collaboration of team members, we still completed it smoothly. In this course, I learned about the various uses of AI, got to know the cutting-edge technologies of AI, enriched my knowledge, and hands-on experienced the functions of various AIs. I tried to use different AIs to provide ideas for system development and continuously optimized our system. The teamwork of the three-person group has made me deeply understand the power of teamwork and the sense of accomplishment of jointly completing a project, enriching my experience and improving my abilities. In future work, I will also rise to the challenge and face the challenges in system development with more confidence.

10.4 Suggestions for Course Improvement

Our group believes that the course design is relatively reasonable. Group members actively participated in the entire system development process, experienced the whole process, and each member has gained a lot. This is mainly reflected in the following aspects:

1. Improvement of team communication and collaboration skills: When working on a group project to develop a system, we often chat and discuss matters, jointly address issues, with each person responsible for a part of the work and then combine them. Through this approach, we learned how to

How to better cooperate, teamwork has become increasingly harmonious.

2. Cultivation of Practical Abilities: The group brainstorming selected real system development projects, deeply integrating theoretical knowledge into practical aspects such as code writing, system development using various AIs, and functional testing. Through multiple

rounds of debugging and optimization, as well as solving practical technical problems, group members not only deepened their understanding of the information system development process and principles but also honed solid hands-on operation and problem-solving abilities.

3. Stimulate innovative thinking: To precisely meet the diverse needs of users, the entire process of system development is permeated with innovative exploration of functional design, interactive experience, and technical solutions. In order to develop a more practical and convenient system, we must constantly try new ideas and methods. Whether it is optimizing the User Interface design or adding featured functional modules, it requires breaking through conventional thinking. Even if some ideas fail to succeed in practice, through repeated exploration and improvement, we gradually develop the habit of active thinking and courage to innovate, and innovative thinking is fully cultivated.

Some feelings of the group members: The time for hands-on practice and assignment submission in the experimental class is relatively tight. After attending the offline part of the experimental class, there is no specific discussion area for the online part, and searching for it every time is rather hasty, which consumes some time.

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